1. **(d)** : Let $\sin^{-1}x=\theta,\ \frac{-\pi}{4}<\theta<\frac{\pi}{4}$

$\Rightarrow\ x=\sin\theta$

Now, $\sin^{-1}\left(\frac{\sqrt{3}}{2}x+\frac{1}{2}\sqrt{1-x^2}\right)$

$=\sin^{-1}\left(\sin\theta\cos\frac{\pi}{6}+\cos\theta\sin\frac{\pi}{6}\right)$

$=\sin^{-1}\left(\sin\left(\theta+\frac{\pi}{6}\right)\right)=\theta+\frac{\pi}{6}\ =\sin^{-1}x+\frac{\pi}{6}$

2. **(b)** : The given infinite series is :

$\cot^{-1}\left(\frac{7}{4}\right)+\cot^{-1}\left(\frac{19}{4}\right)+\cot^{-1}\left(\frac{39}{4}\right)+\cot^{-1}\left(\frac{67}{4}\right)+.....$

$\Rightarrow\ T_n=\cot^{-1}\left(\frac{4n^2+3}{4}\right)$

$=\tan^{-1}\left(\frac{4}{4n^2+3}\right)=\tan^{-1}\left(\frac{1}{n^2+\frac{3}{4}}\right)$

$=\tan^{-1}\left(\frac{1}{n^2-\frac{1}{4}+1}\right)$

$=\tan^{-1}\left(\frac{\left(n+\frac{1}{2}\right)-\left(n-\frac{1}{2}\right)}{1+\left(n+\frac{1}{2}\right)\left(n-\frac{1}{2}\right)}\right)$

$=\tan^{-1}\left(n+\frac{1}{2}\right)-\tan^{-1}\left(n-\frac{1}{2}\right)$

$T_1=\tan^{-1}\frac{3}{2}-\tan^{-1}\frac{1}{2}$

$T_2=\tan^{-1}\frac{5}{2}-\tan^{-1}\frac{3}{2}$ and so on.

$\therefore\quad \sum_{r=1}^{n}T_r=\tan^{-1}\left(n+\frac{1}{2}\right)-\tan^{-1}\frac{1}{2}$

as $n\to\infty$

$\sum_{r=1}^{\infty}T_r=\frac{\pi}{2}-\tan^{-1}\frac{1}{2}$

3. **(c)** : We have

$f(x)=\log_4\log_3\log_7(8-\log_2(x^2+4x+5)$

For $f(x)$ to be defined we need,

$\log_3\log_7(8-\log_2(x^2+4x+5))>0.$

$\Rightarrow\ \log_7(8-\log_2(x^2+4x+5)>1$

$\Rightarrow\ 8-\log_2(x^2+4x+5)>7$

$\Rightarrow\ \log_2(x^2+4x+5)<1$

$\Rightarrow\ x^2+4x+5<2\Rightarrow x^2+4x+3<0$

$\Rightarrow\ (x+1)(x+3)<0$

$\Rightarrow\ x\in(-3,-1)$ *i.e.*, $\alpha=-3,\beta=-1$ ...(i)

Also, $x^2+4x+5>0$, but $D=16-20=-4<0$

Also, we have $g(x)=\sin^{-1}\left(\frac{7x+10}{x-2}\right)$

$\Rightarrow\ -1\le\frac{7x+10}{x-2}\le 1$

$\Rightarrow\ -1\le\frac{7(x-2)+24}{x-2}\le 1$

$\Rightarrow\ -8\le\frac{24}{x-2}\le -6\ \Rightarrow\ \frac{1}{-6}\le\frac{x-2}{24}\le\frac{1}{-8}$

$\Rightarrow\ -4\le x-2\le -3\Rightarrow -2\le x\le -1$

$\Rightarrow\ x\in[-2,-1]$ *i.e.*, $\gamma=-2,\delta=-1$

$\therefore\ \alpha^2+\beta^2+\gamma^2+\delta^2$

$=(-3)^2+(-1)^2+(-2)^2+(-1)^2=15$

4. **(b)** : We have,

$\cot^{-1}\left(\frac{\sqrt{1+\tan^2(2)}-1}{\tan(2)}\right)-\cot^{-1}\left(\frac{\sqrt{1+\tan^2\left(\frac{1}{2}\right)}+1}{\tan\left(\frac{1}{2}\right)}\right)$

$=\cot^{-1}\left(\frac{|\sec(2)|-1}{\tan(2)}\right)-\cot^{-1}\left(\frac{\left|\sec\left(\frac{1}{2}\right)\right|+1}{\tan\left(\frac{1}{2}\right)}\right)$

$=\cot^{-1}\left(\frac{-1-\cos 2}{\sin 2}\right)-\cot^{-1}\left(\frac{1+\cos\left(\frac{1}{2}\right)}{\sin\left(\frac{1}{2}\right)}\right)$

$=\cot^{-1}\left(\frac{-2\ \cos^2(1)}{2\cos(1)\sin(1)}\right)-\cot^{-1}\left(\frac{2\cos^2\left(\frac{1}{4}\right)}{2\cos\left(\frac{1}{4}\right)\sin\left(\frac{1}{4}\right)}\right)$

$=\cot^{-1}(\cot(-1))-\cot^{-1}\left(\cot\frac{1}{4}\right)$

$=\pi-1-\frac{1}{4}=\pi-\frac{5}{4}$

5. **(a)** : $f(x)=16((\sec^{-1}x)^2+(\text{cosec}^{-1}x)^2)$

$=16\,[(\sec^{-1}x+\text{cosec}^{-1}x)^2-2\sec^{-1}x\,\text{cosec}^{-1}x]$

$=16\left[\left(\frac{\pi}{2}\right)^2-2\left(\frac{\pi}{2}-\text{cosec}^{-1}x\right)\text{cosec}^{-1}x\right]$

$=16\left[\frac{\pi^2}{4}-\pi\text{cosec}^{-1}x+2(\text{cosec}^{-1}x)^2\right]$

$=32\left[(\text{cosec}^{-1}x)^2-\frac{\pi}{2}\text{cosec}^{-1}x+\frac{\pi^2}{8}\right]$

$\Rightarrow\ f(x)=32\left[\left(\text{cosec}^{-1}x-\frac{\pi}{4}\right)^2+\frac{\pi^2}{16}\right]$

$f(x)$ is maximum when $x=-1$

$\therefore\quad f_{\max}=32\times 10\frac{\pi^2}{16}$

$f(x)$ is minimum when $x=\sqrt{2}$

$\therefore\quad f_{\min}=32\times\frac{\pi^2}{16}$

$\therefore$ Required sum

$=32\times 10\frac{\pi^2}{16}+32\times\frac{\pi^2}{16}=22\pi^2$

6. **(a)** : Given, $\frac{\pi}{2}\le x\le\frac{3\pi}{4}$

Let $\cos\alpha=\frac{12}{13}$

$\left[\therefore\ \cos^{-1}\left(\frac{12}{13}\cos x+\frac{5}{13}\sin x\right)\right]$

$=\cos^{-1}(\cos x\cos\alpha+\sin x\sin\alpha)$

$=\cos^{-1}(\cos(x-\alpha))=x-\alpha$

$=x-\tan^{-1}\left(\frac{5}{12}\right)$

7. **(d)** : $\beta+\frac{(1+\beta^2)}{\alpha-\beta}=\frac{\alpha\beta-\beta^2+1+\beta^2}{\alpha-\beta}$

$=\frac{\alpha\beta+1}{\alpha-\beta}$

Similarly, $\gamma+\frac{(1+\gamma^2)}{\beta-\gamma}=\frac{\beta\gamma+1}{\beta-\gamma}$

and $\alpha+\frac{(1+\alpha^2)}{\gamma-\alpha}=\frac{\alpha\gamma+1}{\gamma-\alpha}$

$\therefore\quad \cot^{-1}\left(\frac{\alpha\beta+1}{\alpha-\beta}\right)+\cot^{-1}\left(\frac{\beta\gamma+1}{\beta-\gamma}\right)+\cot^{-1}\left(\frac{\alpha\gamma+1}{\gamma-\alpha}\right)$

$=\tan^{-1}\left(\frac{\alpha-\beta}{1+\alpha\beta}\right)+\tan^{-1}\left(\frac{\beta-\gamma}{1-\beta\gamma}\right)+\pi-\cot^{-1}\left(\frac{\alpha\gamma+1}{\alpha-\gamma}\right)$

$=\tan^{-1}\left(\frac{\alpha-\beta}{1+\alpha\beta}\right)+\tan^{-1}\left(\frac{\beta-\gamma}{1+\beta\gamma}\right)-\tan^{-1}\left(\frac{-\gamma+\alpha}{1+\gamma\alpha}\right)+\pi$

$=\tan^{-1}(\alpha)-\tan^{-1}(\beta)+\tan^{-1}(\beta)-\tan^{-1}(\gamma)+\tan^{-1}(\gamma)-\tan^{-1}(\alpha)+\pi=\pi$

8. **(b)** : $\cos\left(\sin^{-1}\frac{3}{5}+\sin^{-1}\frac{5}{13}+\sin^{-1}\frac{33}{65}\right)$

$=\cos\left(\tan^{-1}\frac{3}{4}+\tan^{-1}\frac{5}{12}+\tan^{-1}\frac{33}{56}\right)$

$=\cos\left(\tan^{-1}\frac{\left(\frac{3}{4}+\frac{5}{12}\right)}{\left(1-\frac{3}{4}\cdot\frac{5}{12}\right)}+\tan^{-1}\frac{33}{56}\right)$

$$=\cos\left(\tan^{-1}\frac{56}{33}+\tan^{-1}\frac{33}{56}\right)$$

$$=\cos\left[\cot^{-1}\left(\frac{33}{56}\right)+\tan^{-1}\left(\frac{33}{56}\right)\right]$$

$$=\cos\left(\frac{\pi}{2}\right)=0 \quad \left(\because \quad \tan^{-1}x+\cot^{-1}x=\frac{\pi}{2}\right)$$

9. (a) : We have, $2[x]+1\le -1$

or $2[x]+1\ge 1 \Rightarrow [x]\le -1$ or $[x]\ge 0$

$\Rightarrow$ $x\in(-\infty,0)$ or $x\in[0,\infty)$

$\Rightarrow$ $x\in(-\infty,\infty)$

10. (c) : Clearly, $\theta=0$ satisfy the given equation. So, it is a solution.

Given, $\theta=\tan^{-1}(2\tan\theta)-\frac{1}{2}\sin^{-1}\left(\frac{6\tan\theta}{9+\tan^2\theta}\right)$

Let $\frac{\tan\theta}{3}=\alpha \Rightarrow \tan\theta=3\alpha$

$\Rightarrow$ $\theta=\tan^{-1}(3\alpha)$

So, $\tan^{-1}(3\alpha)=\tan^{-1}(6\alpha)-\frac{1}{2}\sin^{-1}\left(\frac{18\alpha}{9+9\alpha^2}\right)$

$\Rightarrow$ $\tan^{-1}(3\alpha)=\tan^{-1}(6\alpha)-\frac{1}{2}\sin^{-1}\left(\frac{2\alpha}{1+\alpha^2}\right)$

$\Rightarrow$ $\frac{1}{2}\sin^{-1}\left(\frac{2\alpha}{1+\alpha^2}\right)=\tan^{-1}(6\alpha)-\tan^{-1}(3\alpha)$

$\Rightarrow$ $\sin^{-1}\left(\frac{2\alpha}{1+\alpha^2}\right)=2\left\{\tan^{-1}\left(\frac{6\alpha-3\alpha}{1+18\alpha^2}\right)\right\}$

$\Rightarrow$ $\sin^{-1}\left(\frac{2\alpha}{1+\alpha^2}\right)=2\tan^{-1}\left(\frac{3\alpha}{1+18\alpha^2}\right)$

$\Rightarrow$ $\tan^{-1}\left(\frac{2\alpha}{1-\alpha^2}\right)-\tan^{-1}\left(\frac{3\alpha}{1+18\alpha^2}\right)$

$$=\tan^{-1}\left(\frac{3\alpha}{1+18\alpha^2}\right)$$

$\Rightarrow$ $\tan^{-1}\left[\dfrac{\frac{2\alpha}{1-\alpha^2}-\frac{3\alpha}{1+18\alpha^2}}{1+\left(\frac{2\alpha}{1-\alpha^2}\right)\left(\frac{3\alpha}{1+18\alpha^2}\right)}\right]$

$$=\tan^{-1}\left(\frac{3\alpha}{1+18\alpha^2}\right)$$

$\Rightarrow$ $\frac{39\alpha^2-1}{23\alpha^2-18\alpha^4+1}=\frac{3}{1+18\alpha^2}$

Put $\alpha^2=t$,

$$\frac{39t-1}{23t-18t^2+1}=\frac{3}{1+18t}$$

$\Rightarrow$ $756t^2-48t-4=0$

Here, product of roots is negative it means one root is positive and other root is negative but negative root is neglected as $\alpha^2=t$.

Thus, only two real solution exists.

Hence, total number of solution is 3.

11. (b) : $\sin^{-1}\left(\frac{3x-22}{2x-19}\right)+\log_e\left(\frac{3x^2-8x+5}{x^2-3x-10}\right)$

Since, $-1\le\sin^{-1}x\le 1$

$\therefore$ $-1\le\frac{3x-22}{2x-19}\le 1$

$\frac{3x-22}{2x-19}+1\ge 0$ and $\frac{3x-22}{2x-19}-1\le 0$

$\Rightarrow$ $\frac{3x-22+2x-19}{2x-19}\ge 0$ and

$\frac{3x-22-2x+19}{2x-19}\le 0$

$\Rightarrow$ $\frac{5x-41}{2x-19}\ge 0$ and $\frac{x-3}{2x-19}\le 0$

$x\in\left(-\infty,\frac{41}{5}\right]\cup\left(\frac{19}{2},\infty\right)$ and $x\in\left[3,\frac{19}{2}\right)$

$\Rightarrow$ $x\in\left[3,\frac{41}{5}\right]$ ...(i)

and, $\frac{3x^2-8x+5}{x^2-3x-10}>0$; $\frac{(3x-5)(x-1)}{(x-5)(x+2)}>0$

$\Rightarrow$ $x\in(-\infty,-2)\cup\left(1,\frac{5}{3}\right)\cup(5,\infty)$ ...(ii)

Taking intersection of individual domains

$x\in\left(5,\frac{41}{5}\right] \Rightarrow \alpha=5$ and $\beta=\frac{41}{5}$

Hence, $3\alpha+10\beta=15+82=97$

12. (b) : $\cos^{-1}x-\frac{\pi}{2}+\cos^{-1}y=\alpha$

$\Rightarrow$ $\cos^{-1}x+\cos^{-1}y=\frac{\pi}{2}+\alpha$

$\therefore$ $\frac{\pi}{2}+\alpha\in\left(0,\frac{3\pi}{2}\right)$ $\left[\because \alpha\in\left[\frac{-\pi}{2},\pi\right]\right]$

$\cos^{-1}\left(xy-\sqrt{1-x^2}\sqrt{1-y^2}\right)=\frac{\pi}{2}+\alpha$

$\Rightarrow$ $xy-\sqrt{1-x^2}\sqrt{1-y^2}=-\sin\alpha$

$\Rightarrow$ $xy+\sin\alpha=\sqrt{1-x^2}\sqrt{1-y^2}$

$\Rightarrow$ $x^2y^2+\sin^2\alpha+2xy\sin\alpha=1-x^2-y^2+x^2y^2$

$\Rightarrow$ $x^2+y^2+2xy\sin\alpha=\cos^2\alpha$

Thus, required minimum value is 0.

13. (b) : We have, $f(x)=\sin^{-1}\left(\frac{x-1}{2x+3}\right)$, since domain of $\sin^{-1}x$ is $[-1,1]$, $\forall x$.

$\Rightarrow$ $-1\le\frac{x-1}{2x+3}\le 1$

$\Rightarrow$ $\frac{x-1}{2x+3}+1\ge 0$ and $\frac{x-1}{2x+3}-1\le 0$

$\Rightarrow$ $\frac{x-1+2x+3}{2x+3}\ge 0$ and $\frac{x-1-2x-3}{2x+3}\le 0$

$\Rightarrow$ $\frac{3x+2}{2x+3}\ge 0$ and $\frac{-x-4}{2x+3}\le 0$

Now, $\frac{3x+2}{2x+3}\ge 0 \Rightarrow x<-\frac{3}{2}$ or $x\ge -\frac{2}{3}$

$\Rightarrow$ $x\in\left(-\infty,-\frac{3}{2}\right)\cup\left[\frac{-2}{3},\infty\right)$

and $\frac{-x-4}{2x+3}\le 0 \Rightarrow x\le -4$ or $x>-\frac{3}{2}$

$\Rightarrow$ $x\in(-\infty,-4]\cup\left(\frac{-3}{2},\infty\right)$

Hence,

$x\in(-\infty,-4]\cup\left[\frac{-2}{3},\infty\right)$ *i.e.,* $x\in R-\left(-4,\frac{-2}{3}\right)$

So, Domain : $R-\left(-4,\frac{-2}{3}\right)$

$\Rightarrow$ $\alpha=-4$ and $\beta=\frac{-2}{3}$

$\therefore$ $12\times\beta=12\times(-4)\times\left(-\frac{2}{3}\right)=32$

14. (a) : Given, $\tan^{-1}(x)+\tan^{-1}(2x)=\frac{\pi}{4}$

$\Rightarrow$ $\tan^{-1}\left(\frac{3x}{1-2x^2}\right)=\frac{\pi}{4}$

$\left(\because \tan^{-1}x+\tan^{-1}y=\tan^{-1}\left(\frac{x+y}{1-xy}\right)\right)$

$\Rightarrow$ $\frac{3x}{1-2x^2}=\tan\frac{\pi}{4} \Rightarrow \frac{3x}{1-2x^2}=1$

$\Rightarrow$ $2x^2+3x-1=0$

$\Rightarrow$ $x=\frac{-3\pm\sqrt{17}}{4}$

$\therefore$ Possible value of $x=\frac{-3+\sqrt{17}}{4}$

Hence, only 1 positive real value of x satisfying the equation.

15. (d) : We have, $\cos\left(2\sin^{-1}x\right)=\frac{1}{9}$

Put $\sin^{-1}x=\theta$

$\cos2\theta=\frac{1}{9} \Rightarrow 1-2\sin^2\theta=\frac{1}{9}$

$\Rightarrow$ $\sin\theta=\frac{2}{3}$

($\because$ $x>0$ and $\sin\theta$ lies in I Quadrant)

$\Rightarrow$ $x=\frac{2}{3}=\frac{m}{n}$. So, $m=2$ and $n=3$

The given equation becomes

$2x^2-3x-2+3=0$

$\Rightarrow$ $2x^2-3x+1=0$

$\Rightarrow$ $2x^2-2x-x+1=0$

$\Rightarrow$ $(2x-1)(x-1)=0$

$\Rightarrow$ $x=\frac{1}{2}$ or $x=1$

$\Rightarrow$ $\alpha=1$ and $\beta=\frac{1}{2}$, which lies on the line $5x+8y=9$

16. (c) : We have,

$f(x)=\cos^{-1}\left(\frac{2-|x|}{4}\right)+\{\log_e(3-x)\}^{-1}$

For $f(x)$ be defined $-1\le\frac{2-|x|}{4}\le 1$

$\Rightarrow$ $-4\le 2-|x|\le 4 \Rightarrow -6\le -|x|\le 2$

$\Rightarrow$ $6\ge |x|\ge -2$

Since, $|x|\ge -2$ so $-6\le x\le 6$ ...(i)

Also, $3-x>0$ and $3-x\neq 1$

$\Rightarrow \quad x<3$ and $x\neq 2$ ...(ii)

Taking intersection of (i) and (ii), we get

$x\in[-6,3)-\{2\}$

$\Rightarrow \quad \alpha=6,\ \beta=3$ and $\gamma=2$

$\Rightarrow \quad \alpha+\beta+\gamma=6+3+2=11$

17. (a) : Given function,

$$f(x)=\log_e\left(\frac{2x+3}{4x^2+x-3}\right)+\cos^{-1}\left(\frac{2x-1}{x+2}\right)$$

Here, $\log\left(\frac{2x+3}{4x^2+x-3}\right)$ will be defined, if

$$\frac{2x+3}{4x^2+x-3}>0 \Rightarrow \frac{2x+3}{(4x-3)(x+1)}>0$$

$-$ $+$ $-$ $+$

$-\infty$ $\quad -3/2$ $\quad -1$ $\quad 3/4$ $\quad \infty$

$\therefore\ x\in\left(\frac{-3}{2},-1\right)\cup\left(\frac{3}{4},\infty\right)$ ...(i)

Also, $\cos^{-1}\left(\frac{2x-1}{x+2}\right)$ will be defined, if

$$-1\le\frac{2x-1}{x+2}\le 1 \Rightarrow 0\le\frac{3x+1}{x+2}$$

$+$ $-$ $+$

$-\infty$ $\quad -2$ $\quad -1/3$ $\quad \infty$

$\therefore \quad x\in\left(-\infty,-2]\cup\left[\frac{-1}{3},\infty\right]\right)$ ...(ii)

Also, $\frac{x-3}{x+2}\le 0$

$-$ $+$ $-$

$-\infty$ $\quad -2$ $\quad 3$ $\quad \infty$

$\therefore \quad x\in[-2,3]$ ...(iii)

From (i), (ii) & (iii), we get

$\beta=3$ and $\alpha=\frac{3}{4} \Rightarrow 5\beta-4\alpha=12$

18. (a) : Given, $\sin^{-1}\alpha+\sin^{-1}\beta+\sin^{-1}\gamma=\pi$

and $(\alpha+\beta+\gamma)(\alpha+\beta-\gamma)=3\alpha\beta$

Let $\sin^{-1}\alpha=A$, $\sin^{-1}\beta=B$, $\sin^{-1}\gamma=C$

$\Rightarrow \quad A+B+C=\pi$ and $(\alpha+\beta+\gamma)(\alpha+\beta-\gamma)=3\alpha\beta$

$\alpha^2+\beta^2+2\alpha\beta-\gamma^2=3\alpha\beta,\ \alpha^2+\beta^2-\gamma^2=\alpha\beta$

$$\frac{\alpha^2+\beta^2-\gamma^2}{2\alpha\beta}=\frac{1}{2} \Rightarrow \cos C=\frac{1}{2} \Rightarrow C=60^\circ$$

$\Rightarrow \quad \sin C=\frac{\sqrt{3}}{2}=\gamma$ So, $\gamma=\frac{\sqrt{3}}{2}$

19. (b) : We have, $a=\sin^{-1}(\sin(5))$ and $b=\cos^{-1}(\cos(5))$

$\Rightarrow \quad a=-2\pi+5$

$\left(\because \sin^{-1}(\sin x)=-2\pi+x, \text{ if } \frac{3\pi}{2}\le x\le\frac{5\pi}{2}\right)$

$b=2\pi-5,$

$(\because \cos^{-1}(\cos x)=2\pi-x,\ \pi\le x\le 2\pi)$

$\therefore \quad a^2+b^2=(-2\pi+5)^2+(2\pi-5)^2$

$=4\pi^2+25-20\pi+4\pi^2+25-20\pi$

$=8\pi^2-40\pi+50.$

20. (b) : $\sin^{-1}\frac{3}{5}=\tan^{-1}\frac{3}{4}$

$$2\cos^{-1}\frac{2}{\sqrt{5}}=2\tan^{-1}\frac{1}{2}=\tan^{-1}\frac{2\cdot\frac{1}{2}}{1-\frac{1}{4}}=\tan^{-1}\frac{4}{3}$$

Now,

$$\tan\left(\tan^{-1}\frac{3}{4}-\tan^{-1}\frac{4}{3}\right)=\tan\left(\tan^{-1}\frac{\frac{3}{4}-\frac{4}{3}}{1+\frac{3}{4}\cdot\frac{4}{3}}\right)$$

$$=\tan\left(\tan^{-1}\left(\frac{-7}{24}\right)\right)=\frac{-7}{24}$$

21. (*) : We have,

$\frac{6+2\log_3 x}{-5x}>0$ and $x>0$ & $x\neq\frac{1}{3}$

This gives $x\in\left(0,\frac{1}{27}\right)$ ...(i)

$$\Rightarrow \quad -1\le\log_{3x}\left(\frac{6+2\log_3 x}{-5x}\right)\le 1$$

$\left[\because \sin^{-1}\theta\in[-1,1]\right]$

$$\Rightarrow \quad 3x\le\frac{6+2\log_3 x}{-5x}\le\frac{1}{3x}$$

We have two cases such as

$15x^2+6+2\log_{3x}\ge 0$ $\qquad$ $6+2\log_{3x}x+\frac{5}{3}\ge 0$

$x\in(0,\infty)$ ...(2) $\qquad$ $x\ge 3^{-23/6}$ ...(3)

Now from eqn. (1), (2) and (3), we get

$x\in\left(3^{-23/6},\frac{1}{27}\right)$

$\therefore \quad \alpha$ is small positive quantity.

and $\beta=\frac{1}{27}$

$\therefore \quad \alpha^2+\frac{5}{\beta}$ is just greater than 135.

22. (a) : Given, $4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)$

$$\frac{x^2}{x^2+1}=\frac{x^2+1-1}{x^2+1}=1-\frac{1}{x^2+1}$$

$x^2\ge 0;\ x^2+1\ge 1$

$$\therefore \quad 0<\frac{1}{x^2+1}\le 1 \Rightarrow -1\le\frac{-1}{x^2+1}<0;$$

$$0\le 1-\frac{1}{x^2+1}<1;\ 0\le 4\sin^{-1}\left(\frac{x^2}{x^2+1}\right)<2\pi$$

23. (a) : We have,

$$f(x)=\log_e(4x^2+11x+6)+\sin^{-1}(4x+3)+\cos^{-1}\left(\frac{10x+6}{3}\right)$$

Let $f_1(x)=\log_e(4x^2+11x+6)$

$f_2(x)=\sin^{-1}(4x+3)$

and $f_3(x)=\cos^{-1}\left(\frac{10x+6}{3}\right)$

Now, we will find domain of f_1, f_2 and f_3

Consider, $f_1(x)=\log(4x^2+11x+6)$

Now, $4x^2+11x+6>0$

$\Rightarrow \quad x>\frac{-11+\sqrt{121-96}}{8}$ and

$x<\frac{-11-\sqrt{121-96}}{8}$

$\Rightarrow \quad x>\frac{-3}{4}$ and $x<-2$

Consider, $f_2(x)=\sin^{-1}(4x+3)$

So, $-1\le 4x+3\le 1$

$\Rightarrow \quad -4\le 4x\le -2 \Rightarrow -1\le x\le -\frac{1}{2}$

$\therefore \quad D(f_2(x))$ is $\left[-1,-\frac{1}{2}\right]$

Consider, $f_3(x)=\cos^{-1}\left(\frac{10x+6}{3}\right)$

So, $-1\le\frac{10x+6}{3}\le 1$

$\Rightarrow \quad -3\le 10x+6\le 3 \Rightarrow \frac{-9}{10}\le x\le\frac{-3}{10}$

$\Rightarrow \quad D(f_3(x))$ is $\left[\frac{-9}{10},\frac{-3}{10}\right]$

Now, $D(f(x))$ is

$D(f_1(x))\cap D(f_2(x))\cap D(f_3(x))$

$$=(-\infty,-2)\cup\left(\frac{-3}{4},\infty\right)\cap\left[-1,\frac{-1}{2}\right]\cap\left[\frac{-9}{10},\frac{-3}{10}\right]$$

$$=\left(\frac{-3}{4},\frac{-1}{2}\right]$$

So, $\alpha=-\frac{3}{4}$ and $\beta=\frac{-1}{2}$

So, $36|\alpha+\beta|=36\left|-\frac{3}{4}-\frac{1}{2}\right|$

$=36\times\frac{5}{4}=45$

24. (b)

25. (b) : $0<x<1$

$$2\tan^{-1}\left(\frac{1-x}{1+x}\right)=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) \quad \text{...(i)}$$

Let $\tan^{-1}x=\theta\in\left(0,\frac{\pi}{4}\right) \Rightarrow x=\tan\theta$

$$\Rightarrow \quad 2\tan^{-1}\left(\tan\left(\frac{\pi}{4}-\theta\right)\right)=\cos^{-1}(\cos 2\theta)$$

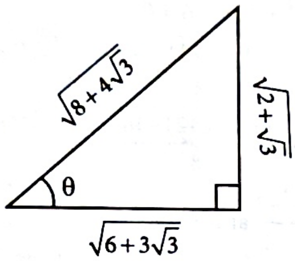
$$\Rightarrow \quad 2\left(\frac{\pi}{4}-\theta\right)=2\theta$$

$\Rightarrow \quad 4\theta=\frac{\pi}{2} \quad \therefore\ \theta=\frac{\pi}{8}$

$x=\tan\frac{\pi}{8} \Rightarrow x=\sqrt{2}-1=0.414$

26. (d) : $\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right)$

$$+\sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$$



$$\tan^{-1}\left(\frac{1+\sqrt{3}}{\sqrt{3}(\sqrt{3}+1)}\right)+\tan^{-1}\left(\frac{\sqrt{2+\sqrt{3}}}{\sqrt{6+3\sqrt{3}}}\right)$$

$$\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)+\tan^{-1}\left(\frac{\sqrt{2+\sqrt{3}}}{\sqrt{3(2+\sqrt{3})}}\right)$$

$$\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)+\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)=\frac{\pi}{6}+\frac{\pi}{6}=\frac{\pi}{3}$$

27. (b, c) : $a_1 = 1, a_2, a_3, a_4, \ldots$ be consecutive natural numbers

$\therefore \quad a_2 = 2, a_3 = 3, \ldots, a_{2021} = 2021, a_{2022} = 2022$

$$\tan^{-1}\left[\frac{1}{1+a_1a_2}\right]=\tan^{-1}\left[\frac{1}{1+1\cdot 2}\right]=\tan^{-1}(1)-\tan^{-1}\left(\frac{1}{2}\right)$$

$$\tan^{-1}\left[\frac{1}{1+a_2a_3}\right]=\tan^{-1}\left[\frac{1}{1+2\cdot 3}\right]=\tan^{-1}\left(\frac{1}{2}\right)-\tan^{-1}\left(\frac{1}{3}\right)$$

$$\tan^{-1}\left[\frac{1}{1+a_{2021}a_{2022}}\right]=\tan^{-1}\left[\frac{1}{1+2021\cdot 2022}\right]$$

$$=\tan^{-1}\left(\frac{1}{2021}\right)-\tan^{-1}\left(\frac{1}{2022}\right)$$

$$\therefore\ \tan^{-1}\left(\frac{1}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{1}{1+a_2a_3}\right)+\ldots\ldots+\tan^{-1}\left(\frac{1}{1+a_{2021}a_{2022}}\right)$$

$$=\tan^{-1}(1)-\tan^{-1}\left(\frac{1}{2022}\right)=\frac{\pi}{4}-\cot^{-1}(2022)$$

$$=\frac{\pi}{4}-\left(\frac{\pi}{2}-\tan^{-1}(2022)\right)=\tan^{-1}(2022)-\frac{\pi}{4}$$

28. (d) : We have,

$$\sin^{-1}\frac{\alpha}{17}+\cos^{-1}\frac{4}{5}-\tan^{-1}\frac{77}{36}=0;\ 0<\alpha<13$$

$$\Rightarrow \sin^{-1}\frac{\alpha}{17}=\tan^{-1}\frac{77}{36}-\cos^{-1}\frac{4}{5}=\tan^{-1}\frac{77}{36}-\sin^{-1}\frac{3}{5}$$

$$\Rightarrow \frac{\alpha}{17}=\sin\left(\tan^{-1}\frac{77}{36}-\sin^{-1}\frac{3}{5}\right)$$

$$=\sin\left(\tan^{-1}\frac{77}{36}\right)\cdot\cos\left(\sin^{-1}\frac{3}{5}\right)-\cos\left(\tan^{-1}\frac{77}{36}\right)\sin\left(\sin^{-1}\frac{3}{5}\right)$$

$$\Rightarrow \frac{\alpha}{17}=\sin\left(\sin^{-1}\frac{77}{85}\right)\cdot\cos\left(\cos^{-1}\frac{4}{5}\right)-\cos\left(\cos^{-1}\frac{36}{85}\right)\left(\frac{3}{5}\right)$$

$$=\frac{77}{85}\times\frac{4}{5}-\frac{36}{85}\times\frac{3}{5}\ \Rightarrow \alpha=8$$

Now, $\sin^{-1}(\sin\alpha)+\cos^{-1}(\cos\alpha)=\sin^{-1}(\sin 8)+\cos^{-1}(\cos 8)$

$$=3\pi-8+8-2\pi\ \left[\because\ 3\pi-8\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]\right]=\pi$$

29. (d) : We know that

$$\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$$

$$\sin^{-1}(\sin\theta)-\left(\frac{\pi}{2}-\sin^{-1}(\sin\theta)\right)>0;$$

This equation hold true if, $\sin^{-1}(\sin\theta)>\frac{\pi}{4}$

$$\Rightarrow \sin\theta>\frac{1}{\sqrt{2}}\quad \text{So, } \theta\in\left(\frac{\pi}{4},\frac{3\pi}{4}\right)$$

$$\theta\in\left(\frac{\pi}{4},\frac{3\pi}{4}\right)=(a,b)$$

$$\Rightarrow b-a=\frac{\pi}{2}=\alpha-\beta\ \text{[Given]}$$

$$\Rightarrow \beta=\alpha-\frac{\pi}{2}$$

Now, $\alpha x^2+\beta x+$

$$\sin^{-1}\left[(x-3)^2+1\right]+\cos^{-1}\left[(x-3)^2+1\right]=0$$

$$\sin^{-1}\left[(x-3)^2+1\right]+\cos^{-1}\left[(x-3)^2+1\right]=\frac{\pi}{2}\text{ if } x=3$$

Put $x=3$, $9\alpha+3\beta+\frac{\pi}{2}=0$; Put $\beta=\alpha-\frac{\pi}{2}$

$$\Rightarrow 9\alpha+3\left(\alpha-\frac{\pi}{2}\right)+\frac{\pi}{2}=0\ \Rightarrow 12\alpha-\pi=0$$

$$\Rightarrow \alpha=\frac{\pi}{12}$$

30. (c) : Case I : $y\in(-3,0)$

$$\therefore\ \tan^{-1}\left(\frac{6y}{9-y^2}\right)+\pi+\tan^{-1}\left(\frac{6y}{9-y^2}\right)=\frac{2\pi}{3}$$

$$\left[\because\ \tan^{-1}\left(\frac{1}{x}\right)=-\pi+\cot^{-1}x,\ x<0\right]$$

$$\Rightarrow 2\tan^{-1}\left(\frac{6y}{9-y^2}\right)=-\frac{\pi}{3}$$

$$\Rightarrow y^2-6\sqrt{3}y-9=0\ \Rightarrow y=3\sqrt{3}-6\quad (\because\ y\in(-3,0))$$

Case II : $y\in(0,3)$

$$\therefore\ 2\tan^{-1}\left(\frac{6y}{9-y^2}\right)=\frac{2\pi}{3}$$

$$\Rightarrow \sqrt{3}y^2+6y-9\sqrt{3}=0$$

$\Rightarrow y=\sqrt{3}$ or $y=-3\sqrt{3}$ (rejected)

$\therefore$ Sum of solutions

$=\sqrt{3}+3\sqrt{3}-6=4\sqrt{3}-6$

31. (b)

32. (b) : $\because\ \dfrac{\sin^{-1}x}{\alpha}=\dfrac{\cos^{-1}x}{\beta}$

$$\Rightarrow \frac{\beta}{\alpha}=\frac{\cos^{-1}x}{\sin^{-1}x}\quad \ldots(i)$$

Now, $\sin\left(\dfrac{2\pi\alpha}{\alpha+\beta}\right)=\sin\left(\dfrac{2\pi}{1+\beta/\alpha}\right)$

$$=\sin\left(\frac{2\pi\sin^{-1}x}{\pi/2}\right)=\sin(4\sin^{-1}x)$$

$$=\sin\left(2\sin^{-1}\left(2x\sqrt{(1-x^2)}\right)\right)$$

$$=\sin\left(\sin^{-1}\left[\left(2\cdot 2x\sqrt{(1-x^2)}\right)\times\left(\sqrt{1-4x^2(1-x^2)}\right)\right]\right)$$

$$=4x\sqrt{(1-x^2)}\ \sqrt{1-4x^2+4x^4}$$

$$=4x\sqrt{(1-x^2)}\cdot(1-2x^2)$$

33. (b) : For domain,

$$-1\le\frac{x^2-4x+2}{x^2+3}\le 1$$

Case-I : $x^2-4x+2\ge -x^2-3$

$\Rightarrow \quad 2x^2-4x+5\ge 0\ \Rightarrow\ x\in(-\infty,\infty)$

Case-II : $x^2-4x+2\le x^2+3$

$$\Rightarrow \quad -4x+2\le 3\ \Rightarrow\ x\ge\frac{-1}{4}\ \therefore\ x\in\left[\frac{-1}{4},\infty\right)$$

So, $D_f=\left[\dfrac{-1}{4},\infty\right)$

34. (a) : We have, $\cos^{-1}x-2\sin^{-1}x=\cos^{-1}2x$

$\Rightarrow \quad \dfrac{\pi}{2}-3\sin^{-1}x=\dfrac{\pi}{2}-\sin^{-1}2x$

$\Rightarrow \quad 3\sin^{-1}x=\sin^{-1}2x$

$\Rightarrow \quad \sin^{-1}(3x-4x^3)=\sin^{-1}2x$

$\Rightarrow \quad 3x-4x^3=2x\ \Rightarrow\ x(1-4x^2)=0$

$\Rightarrow \quad x=0,\dfrac{-1}{2},\dfrac{1}{2}$

$\therefore \quad$ Required sum $=0$

35. (c) : Consider

$$f(x)=\sin^{-1}\left(\frac{x^2-3x+2}{x^2+2x+7}\right),$$

where $-1\le\dfrac{x^2-3x+2}{x^2+2x+7}\le 1$

$\Rightarrow \quad -x^2-2x-7\le x^2-3x+2\le x^2+2x+7$

$\Rightarrow \quad 0\le 2x^2-x+9$ and $5x+5\ge 0$

$\Rightarrow \quad 2x^2-x+9\ge 0\Rightarrow x\in R$ and $x+1\ge 0$

$\therefore \quad$ Domain of $f(x)=[-1,\infty)$

36. (a) : We have,

$(\tan^{-1}x)^3+(\cot^{-1}x)^3=k\pi^3 \quad \ldots(i)$

Let $y=(\tan^{-1}x)^3+(\cot^{-1}x)^3$

$= (\tan^{-1}x + \cot^{-1}x)\,[(\tan^{-1} x)^2 + (\cot^{-1}x)^2 - \tan^{-1}x \cdot \cot^{-1} x]$

$$=\frac{\pi}{2}\left[t^2+\left(\frac{\pi}{2}-t\right)^2-t\left(\frac{\pi}{2}-t\right)\right]$$

[where $t = \tan^{-1} x$]

$$=\frac{\pi}{2}\left[3t^2-\frac{3}{2}\pi t+\frac{\pi^2}{4}\right]$$

Now, for maximum/minimum, put $y'(t) = 0$

$$\Rightarrow \quad 6t-\frac{3\pi}{2}=0 \Rightarrow t=\frac{\pi}{4}$$

So, $y_{\min}$ at $t=\frac{\pi}{4}$ and $y_{\max}$ at $t=-\frac{\pi}{2}$

$$\therefore \quad y_{\min}=\frac{\pi}{2}\left[\frac{3\pi^2}{16}-\frac{3\pi^2}{8}+\frac{\pi^2}{4}\right]=\frac{\pi^3}{32}$$

and $y_{\max}=\frac{\pi}{2}\left[\frac{3\pi^2}{4}+\frac{3\pi^2}{4}+\frac{\pi^2}{4}\right]=\frac{7\pi^3}{8}$

From (i), we have,

$$\frac{\pi^3}{32}\le k\pi^3<\frac{7}{8}\pi^3 \Rightarrow \frac{1}{32}\le k<\frac{7}{8}$$

$$\Rightarrow \quad k\in\left[\frac{1}{32},\frac{7}{8}\right)$$

37. (b) : We have, $x * y = x^2 + y^3$.

Also, $(x * 1) * 1 = x * (1 * 1)$

$\Rightarrow \quad (x^2 + 1) * 1 = x * (1 + 1)$

$\Rightarrow \quad (x^2 + 1)^2 + 1 = x^2 + 2^3$

$\Rightarrow \quad x^4 + 2x^2 + 1 + 1 = x^2 + 8$

$\Rightarrow \quad x^4 + x^2 = 6$ ...(i)

Now, $2\sin^{-1}\left(\frac{x^4+x^2-2}{x^4+x^2+2}\right)$

$=2\sin^{-2}\left(\frac{6-2}{6+2}\right)$ (Using (i))

$$=2\sin^{-1}\left(\frac{1}{2}\right)=\frac{2\pi}{6}=\frac{\pi}{3}$$

38. (b) : $\tan^{-1}\left[\frac{\cos(15\pi/4)-1}{\sin(\pi/4)}\right]$

$$=\tan^{-1}\left[\frac{\cos\left(4\pi-\frac{\pi}{4}\right)-1}{\frac{1}{\sqrt{2}}}\right]$$

$$=\tan^{-1}\left[\sqrt{2}\left(\frac{1}{\sqrt{2}}-1\right)\right]=\tan^{-1}\left(1-\sqrt{2}\right)$$

$$=\tan^{-1}\left[\tan\left(\frac{-\pi}{8}\right)\right]=\frac{-\pi}{8}$$

39. (c)

40. (a) : $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)+\cos^{-1}\left(\cos\frac{7\pi}{6}\right)+\tan^{-1}\left(\tan\frac{3\pi}{4}\right)$

$$=\sin^{-1}\left(\sin\left(\pi-\frac{\pi}{3}\right)\right)+\cos^{-1}\left(\cos\left(\pi+\frac{\pi}{6}\right)\right)+\tan^{-1}\left(\tan\left(\pi-\frac{\pi}{4}\right)\right)$$

$$=\frac{\pi}{3}+(\pi-\pi/6)-\frac{\pi}{4}=\frac{\pi}{3}+\frac{5\pi}{6}-\frac{\pi}{4}=\frac{11\pi}{12}$$

41. (a) : $\cot\left(\sum_{n=1}^{50}\tan^{-1}\left(\frac{1}{1+n+n^2}\right)\right)$

$$=\cot\left(\sum_{n=1}^{50}\tan^{-1}\left(\frac{(n+1)-n}{1+n(n+1)}\right)\right)$$

$$=\cot\left(\sum_{n=1}^{50}\{\tan^{-1}(n+1)-\tan^{-1}n\}\right)$$

$$=\cot(\tan^{-1}(51)-\tan^{-1}(1))=\cot\left(\tan^{-1}\left(\frac{51-1}{1+51}\right)\right)$$

$$=\cot\left(\tan^{-1}\left(\frac{50}{52}\right)\right)=\frac{52}{50}=\frac{26}{25}$$

42. (d) : $y=\cos^{-1}\left(\frac{2\sin^{-1}(1/(4x^2-1))}{\pi}\right)$

For domain, $-1\le\frac{1}{4x^2-1}\le1$

$\Rightarrow \quad 4x^2 - 1 \ge 1$ or $4x^2 - 1 \le -1$ and $4x^2 - 1 \ne 0$

$\Rightarrow \quad x^2\ge\frac{1}{2}$ or $4x^2 \le 0$ and $x\ne\pm\frac{1}{2}$

$$\Rightarrow \quad \left(x-\frac{1}{\sqrt{2}}\right)\left(x+\frac{1}{\sqrt{2}}\right)\ge0 \text{ or } x=0$$

$$\Rightarrow \quad x\in\left(-\infty,\frac{-1}{\sqrt{2}}\right]\cup\left[\frac{1}{\sqrt{2}},\infty\right)\cup\{0\}$$

Now, for the given x,

$$\frac{2}{\pi}\sin^{-1}\left(\frac{1}{4x^2-1}\right)\in(0,1)\cup\{-1\}$$

Hence, domain of

$$y=\left(-\infty,\frac{-1}{\sqrt{2}}\right]\cup\left[\frac{1}{\sqrt{2}},\infty\right)\cup\{0\}$$

43. (c) : $\cos^{-1}(\cos(-5))$

$= \cos^{-1}(\cos(2\pi - 5)) = 2\pi - 5$

$\sin^{-1}(\sin(6)) = \sin^{-1}[\sin(-(2\pi - 6))] = 6 - 2\pi$

$\tan^{-1}(\tan(12)) = \tan^{-1}(\tan(12 - 4\pi)) = 12 - 4\pi$

Now, $\cos^{-1}(\cos(-5))+\sin^{-1}(\sin(6))-\tan^{-1}(\tan(12))$

$= 2\pi - 5 + 6 - 2\pi - (12 - 4\pi) = 4\pi - 11$

44. (c)

45. (a) : $\text{cosec}^{-1}\left(\frac{1+x}{x}\right)$ is defined

if $\left|\frac{1+x}{x}\right|\ge1$

$$\Rightarrow \quad \frac{(1+x)^2}{x^2}\ge1 \Rightarrow 1+x^2+2x\ge x^2 \Rightarrow x\ge\frac{-1}{2}$$

Also, for $x = 0$, $\text{cosec}^{-1}\left(\frac{1+x}{x}\right)$ is undefined.

Domain of the function is $x\in\left[\frac{-1}{2},\infty\right)-\{0\}$

46. (a) **47. (b)**

48. (b) : We have, $f(x) = \tan^{-1}(\sin x + \cos x)$

Here, $1 \le \sin x + \cos x \le \sqrt{2}\ \forall\, x\in\left[0,\frac{\pi}{2}\right]$

$M = \tan^{-1}(\sqrt{2})$ and $m = \tan^{-1}(1)$

$$\therefore \quad M-m=\tan^{-1}\left(\frac{\sqrt{2}-1}{1+\sqrt{2}}\right)=\tan^{-1}(3-2\sqrt{2})$$

Hence, $\tan(M - m) = 3 - 2\sqrt{2}$

49. (c)

50. (b) : We have,

$$\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{x^2+x+1}=\frac{\pi}{4} \quad \text{...(i)}$$

The equation (i) is defined for

$x^2 + x \ge 0$ and $0 \le x^2 + x + 1 \le 1$

$\Rightarrow \quad x \ge 0$ & $x \le -1$ and $-1 \le x^2 + x \le 0$

$\Rightarrow \quad x = 0, -1$ is the only solution but it does not satisfy (i).

$\Rightarrow$ The number of real roots of (i) is zero.

51. (b) : Let $x=2\tan^{-1}\left(\frac{3}{5}\right)+\sin^{-1}\left(\frac{5}{13}\right)$

$$\Rightarrow \quad x=\tan^{-1}\frac{\frac{6}{5}}{1-\frac{9}{25}}+\tan^{-1}\left(\frac{5}{12}\right)$$

$$\Rightarrow \quad x=\tan^{-1}\left(\frac{15}{8}\right)+\tan^{-1}\frac{5}{12}$$

$$=\tan^{-1}\frac{\frac{15}{8}+\frac{5}{12}}{1-\frac{15}{8}\times\frac{5}{12}}=\tan^{-1}\left(\frac{220}{21}\right)$$

$$\therefore \quad \tan(x)=\tan\left(\tan^{-1}\frac{220}{21}\right)=\frac{220}{21}.$$

52. (d) : We have, $f(x)=\frac{\cos^{-1}\sqrt{x^2-x+1}}{\sqrt{\sin^{-1}\left(\frac{2x-1}{2}\right)}}$.

Clearly, $f(x)$ will be well-defined

if $1 \ge x^2 - x + 1 \ge 0$ and $0<\frac{2x-1}{2}\le1$

$\Rightarrow \quad x(x - 1) \le 0$ and $1 < 2x \le 3$

$\Rightarrow \quad x\in[0,1]\cap x\in\left(\frac{1}{2},\frac{3}{2}\right]$

$\Rightarrow \quad x\in[0,1]\cap\left(\frac{1}{2},\frac{3}{2}\right] \Rightarrow x\in\left(\frac{1}{2},1\right]$

$\Rightarrow \quad \alpha = 1/2$ and $\beta = 1$ $\therefore$ $\alpha + \beta = 3/2$.

53. (a) **54. (d)**

55. (a) : We have,

$$\cot^{-1}\alpha=\tan^{-1}\left(\frac{1}{2}\right)+\tan^{-1}\left(\frac{1}{8}\right)+\tan^{-1}\left(\frac{1}{18}\right)+\text{......... upto 100 terms}$$

$$=\sum_{k=1}^{100}\tan^{-1}\left(\frac{1}{2k^2}\right)=\sum_{k=1}^{100}\tan^{-1}\left(\frac{2}{4k^2}\right)$$

$$=\sum_{k=1}^{100}\tan^{-1}\left(\frac{(2k+1)-(2k-1)}{1+(2k-1)(2k+1)}\right)$$

$$=\sum_{k=1}^{100}\left(\tan^{-1}(2k+1)-\tan^{-1}(2k-1)\right)$$

$= \tan^{-1} 201 - \tan^{-1} 1$

$$=\tan^{-1}\left(\frac{201-1}{1+201\times1}\right)=\tan^{-1}\left(\frac{200}{202}\right)$$

$$=\cot^{-1}\left(\frac{101}{100}\right) \Rightarrow \alpha=\frac{101}{100} \text{ or } 1.01.$$

56. (d) : We have,

$\tan^{-1}(x+1)+\cot^{-1}\left(\frac{1}{x-1}\right)$

$=\tan^{-1}(x+1)+\tan^{-1}(x-1)$

$=\tan^{-1}\left(\frac{(x+1)+(x-1)}{1-((x+1)(x-1))}\right)$

$=\tan^{-1}\left(\frac{2x}{2-x^2}\right)=\tan^{-1}\frac{8}{31}$

$\Rightarrow \frac{2x}{2-x^2}=\frac{8}{31}\Rightarrow 4x^2+31x-8=0$

$\Rightarrow x=\frac{1}{4},-8$

But $x = 1/4$ does not satisfy the given equation.

So, sum of possible values of $x=-8=\frac{-32}{4}$

57. (b) : $\sin^{-1}\left[x^2+\frac{1}{3}\right]$ is defined

if $-1\le x^2+\frac{1}{3}<2$

$\Rightarrow 0\le x^2<5/3$ ...(i)

and $\cos^{-1}\left[x^2-\frac{2}{3}\right]$ is defined

if $-1\le x^2-2/3<2$

$\Rightarrow 0\le x^2<8/3$ ...(ii)

From (i) and (ii), $0\le x^2<5/3$

Case I : $0\le x^2<2/3$,

then $\sin^{-1}(0)+\cos^{-1}(-1)=x^2\Rightarrow 0+\pi=x^2$

But $\pi\notin\left[0,\frac{2}{3}\right)\Rightarrow$ No value of x exists.

Case II : If $\frac{2}{3}\le x^2<\frac{5}{3}$,

then $\sin^{-1}(1)+\cos^{-1}(0)=x^2$

$\Rightarrow \frac{\pi}{2}+\frac{\pi}{2}=x^2\Rightarrow x^2=\pi$

But $\pi\notin\left[\frac{2}{3},\frac{5}{3}\right)\Rightarrow$ No value of x exists.

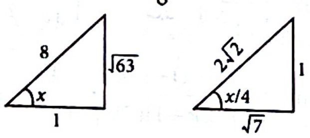
Thus, there exists no solution for the given equation.

58. (d) : $\operatorname{cosec}^{-1}x$ defined for $x\in(-\infty,-1]\cup[1,\infty)$

Also, $\sqrt{\{x\}}>0\Rightarrow x$ can't be integer.

$\therefore$ $f(x)$ is defined for all non-integers except the interval $[-1, 1]$.

59. (d) : Let $\sin^{-1}\frac{\sqrt{63}}{8}=x\Rightarrow \sin x=\frac{\sqrt{63}}{8}$



$\therefore \cos x=\frac{1}{8}$ and $\tan x=\sqrt{63}$

Now, $2\cos^2\frac{x}{2}=1+\cos x=1+\frac{1}{8}=\frac{9}{8}$

$\Rightarrow \cos\frac{x}{2}=\frac{3}{4}$

Also, $2\cos^2\frac{x}{4}=1+\cos\frac{x}{2}=1+\frac{3}{4}=\frac{7}{4}$

$\Rightarrow \cos\frac{x}{4}=\frac{\sqrt{7}}{2\sqrt{2}}$

Now, $\tan\left(\frac{1}{4}\sin^{-1}\frac{\sqrt{63}}{8}\right)=\tan\left(\frac{x}{4}\right)=\frac{1}{\sqrt{7}}$

60. (b) : $\operatorname{cosec}\left[2\tan^{-1}\left(\frac{1}{5}\right)+\tan^{-1}\left(\frac{3}{4}\right)\right]$

$=\operatorname{cosec}\left[\tan^{-1}\left(\frac{2/5}{1-(1/5)^2}\right)+\tan^{-1}\left(\frac{3}{4}\right)\right]$

$=\operatorname{cosec}\left[\tan^{-1}\left(\frac{5}{12}\right)+\tan^{-1}\left(\frac{3}{4}\right)\right]$

$=\operatorname{cosec}\left[\tan^{-1}\left(\frac{\frac{5}{12}+\frac{3}{4}}{1-\frac{5}{12}\cdot\frac{3}{4}}\right)\right]$

$\left[\because \tan^{-1}x+\tan^{-1}y=\tan^{-1}\left(\frac{x+y}{1-xy}\right), xy<1\right]$

$=\operatorname{cosec}\left[\tan^{-1}\left(\frac{56}{33}\right)\right]=\frac{65}{56}$

61. (b) : Let $\frac{\sin^{-1}x}{a}=\frac{\cos^{-1}x}{b}$

$=\frac{\tan^{-1}y}{c}=\lambda\,(\text{say})$

$\Rightarrow \sin^{-1}x=a\lambda,\ \cos^{-1}x=b\lambda,\ \tan^{-1}y=c\lambda$

$\therefore (a+b)\lambda=\sin^{-1}x+\cos^{-1}x=\frac{\pi}{2}$

$\Rightarrow \frac{\pi}{a+b}=2\lambda$

Now, $\cos\left(\frac{\pi c}{a+b}\right)=\cos(2\lambda c)=\cos(2\tan^{-1}y)$

Let $\tan^{-1}y=\theta\Rightarrow y=\tan\theta$

$\therefore \cos\left(\frac{\pi c}{a+b}\right)=\cos 2\theta=\frac{1-\tan^2\theta}{1+\tan^2\theta}=\frac{1-y^2}{1+y^2}$

62. (c) : Given, $\tan^{-1}a+\tan^{-1}b$

$=\frac{\pi}{4}\ \forall\ 0<a,b<1\Rightarrow \frac{a+b}{1-ab}=1$

$\Rightarrow a+b=1-ab\Rightarrow (a+1)(b+1)=2$

Now, $\left[a-\frac{a^2}{2}+\frac{a^3}{3}-.....\right]+\left[b-\frac{b^2}{2}+\frac{b^3}{3}-....\right]$

$=\log_e(1+a)+\log_e(1+b)$

$=\log_e[(1+a)(1+b)]=\log_e 2$

63. (c) : Here, $f(x)=\sin^{-1}\left(\frac{|x|+5}{x^2+1}\right)$

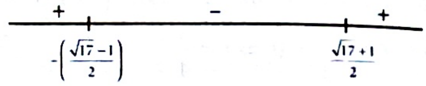
$\Rightarrow \left|\frac{|x|+5}{x^2+1}\right|\le 1\Rightarrow |x|+5\le x^2+1$

$\Rightarrow x^2-4-|x|\ge 0$

$\Rightarrow \left(|x|-\frac{1}{2}\right)^2-\frac{17}{4}\ge 0$

$\Rightarrow \left(|x|-\frac{1}{2}\right)^2-\left(\frac{\sqrt{17}}{2}\right)^2\ge 0$

$\Rightarrow \left(|x|+\frac{\sqrt{17}-1}{2}\right)\left(|x|-\left(\frac{\sqrt{17}+1}{2}\right)\right)\ge 0$



But $|x|$ can't be negative, therefore

$|x|\ge\frac{\sqrt{17}+1}{2}$

$\Rightarrow x\in\left(-\infty,-\left(\frac{\sqrt{17}+1}{2}\right)\right]\cup\left[\frac{\sqrt{17}+1}{2},\infty\right)$

Thus $a=\frac{\sqrt{17}+1}{2}$

[$\because$ Domain of $f(x)$ is $(-\infty,-a]\cup[a,\infty)$]

64. (c) : We have,

$2\pi-\left(\sin^{-1}\frac{4}{5}+\sin^{-1}\frac{5}{13}+\sin^{-1}\frac{16}{65}\right)$

$=2\pi-\left(\sin^{-1}\left(\frac{4}{5}\sqrt{1-\frac{25}{169}}+\frac{5}{13}\sqrt{1-\frac{16}{25}}\right)+\sin^{-1}\frac{16}{65}\right)$

$=2\pi-\left(\sin^{-1}\frac{63}{65}+\sin^{-1}\frac{16}{65}\right)$

$=2\pi-\sin^{-1}\left(\frac{63}{65}\sqrt{1-\frac{256}{4225}}+\frac{16}{65}\sqrt{1-\frac{3969}{4225}}\right)$

$=2\pi-\sin^{-1}\left(\frac{63}{65}\times\frac{63}{65}+\frac{16}{65}\times\frac{16}{65}\right)$

$=2\pi-\sin^{-1}(1)=2\pi-\frac{\pi}{2}=\frac{3\pi}{2}$

65. (a)

66. (b) : Here $\alpha=\cos^{-1}\left(\frac{3}{5}\right)$

and $\beta=\tan^{-1}\left(\frac{1}{3}\right)$

$\Rightarrow \cos\alpha=3/5$ and $\tan\beta=1/3$

$\Rightarrow \sin\alpha=\frac{4}{5}$, $\sin\beta=\frac{1}{\sqrt{10}}$, $\cos\beta=\frac{3}{\sqrt{10}}$

Now, $\sin(\alpha-\beta)=\sin\alpha\cos\beta-\cos\alpha\sin\beta$

$=\frac{4}{5}\times\frac{3}{\sqrt{10}}-\frac{3}{5}\times\frac{1}{\sqrt{10}}=\frac{9}{5\sqrt{10}}$

$\Rightarrow \alpha-\beta=\sin^{-1}\left(\frac{9}{5\sqrt{10}}\right)$

67. (a) : Given, $\cos^{-1}x-\cos^{-1}\frac{y}{2}=\alpha$

$\Rightarrow \cos^{-1}\left\{\frac{xy}{2}+\sqrt{1-x^2}\sqrt{1-\frac{y^2}{4}}\right\}=\alpha$

$\Rightarrow \frac{xy}{2}+\frac{\sqrt{1-x^2}\sqrt{4-y^2}}{2}=\cos\alpha$

$\Rightarrow xy+\sqrt{1-x^2}\sqrt{4-y^2}=2\cos\alpha$

$\Rightarrow \sqrt{1-x^2}\sqrt{4-y^2}=2\cos\alpha-xy$

Squaring both sides, we get

$(1-x^2)(4-y^2)=4\cos^2\alpha+x^2y^2-4xy\cos\alpha$

$\Rightarrow 4-y^2-4x^2+x^2y^2=4\cos^2\alpha+x^2y^2-4xy\cos\alpha$

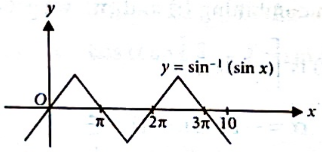
$\Rightarrow 4x^2-4xy\cos\alpha+y^2=4-4\cos^2\alpha$

$\Rightarrow 4x^2-4xy\cos\alpha+y^2=4(1-\cos^2\alpha)$

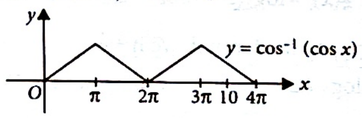
$\Rightarrow 4x^2-4xy\cos\alpha+y^2=4\sin^2\alpha$

68. (a) **69.** (a)

70. (c) :



$x=\sin^{-1}(\sin 10)=3\pi-10$



$y=\cos^{-1}(\cos 10)=4\pi-10$

Now, $y-x=4\pi-10-3\pi+10=\pi$

71. (d)

72. (d) : Here, $(\cot^{-1}x)^2-7(\cot^{-1}x)+10>0$

$\Rightarrow (\cot^{-1}x-5)(\cot^{-1}x-2)>0$

$\Rightarrow \cot^{-1}x>5$ or $\cot^{-1}x<2$

$\Rightarrow x<\cot 5$ or $x>\cot 2$

Since, $x<\cot 5$ does not satisfy the given inequality.

$\therefore\ x\in(\cot 2,\infty)$

73. (d) : Here,

$A=\left\{x\geq 0:\tan^{-1}(2x)+\tan^{-1}(3x)=\frac{\pi}{4}\right\}$

Now, $\tan^{-1}(2x)+\tan^{-1}(3x)=\frac{\pi}{4}$

$\Rightarrow \tan^{-1}\left(\frac{2x+3x}{1-2x\times 3x}\right)=\frac{\pi}{4}$

$\Rightarrow \tan^{-1}\left(\frac{5x}{1-6x^2}\right)=\frac{\pi}{4}$

$\Rightarrow \frac{5x}{1-6x^2}=\tan\frac{\pi}{4}=1$

$\Rightarrow 5x=1-6x^2 \Rightarrow 6x^2+5x-1=0$

$\Rightarrow (6x-1)(x+1)=0$

$\Rightarrow x=\frac{1}{6}$ $[\because x\geq 0]$

74. (b) **75.** (d)

76. (c) : As $|x|<\frac{1}{\sqrt{3}}$, in this range using principal branch of tangent function, we have

$3\tan^{-1}x=\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$

Also, $\tan^{-1}\left(\frac{2x}{1-x^2}\right)=2\tan^{-1}x$

Thus, $\tan^{-1}y=\tan^{-1}x+2\tan^{-1}x=3\tan^{-1}x$

$=\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)\ \therefore\ y=\frac{3x-x^3}{1-3x^2}$

77. (b) : We have, $f(x)$

$=2\tan^{-1}x+\sin^{-1}\left(\frac{2x}{1+x^2}\right)$

$=2\tan^{-1}x+\pi-2\tan^{-1}x\Rightarrow f(x)=\pi$. So, $f(5)=\pi$

78. (d)

79. (b) : $\cot\sum_{n=1}^{23}\left(\cot^{-1}\left(1+\sum_{k=1}^{n}2k\right)\right)$

$=\cot\sum_{n=1}^{23}\cot^{-1}\left(1+\frac{2(n+1)n}{2}\right)$

$=\cot\sum_{n=1}^{23}\cot^{-1}(1+n(n+1))$

$=\cot\sum_{n=1}^{23}\tan^{-1}\left(\frac{1}{1+n(n+1)}\right)$

$=\cot(\tan^{-1}24-\tan^{-1}1)$

$=\cot\left(\tan^{-1}\left(\frac{24-1}{1+24}\right)\right)=\cot\left(\tan^{-1}\frac{23}{25}\right)$

$=\cot\left(\cot^{-1}\frac{25}{23}\right)=\frac{25}{23}$

80. (b) : $\cot\left(\operatorname{cosec}^{-1}\frac{5}{3}+\tan^{-1}\frac{2}{3}\right)$

$=\cot\left(\tan^{-1}\frac{3}{4}+\tan^{-1}\frac{2}{3}\right)$

$=\cot\left(\tan^{-1}\frac{\frac{3}{4}+\frac{2}{3}}{1-\frac{3}{4}\cdot\frac{2}{3}}\right)=\cot\left(\tan^{-1}\frac{17}{6}\right)=\frac{6}{17}$

81. (c) **82.** (d)

83. (b) : $f(x)$ is defined if

$-1\leq\frac{x}{2}-1\leq 1$ and $\cos x>0$

or $0\leq x\leq 4$ and $-\frac{\pi}{2}<x<\frac{\pi}{2}\ \therefore\ 0\leq x<\frac{\pi}{2}$.

84. (d)

85. (d) : $\sin(\cot^{-1}(x+1))$

$=\sin\left(\sin^{-1}\frac{1}{\sqrt{x^2+2x+2}}\right)=\frac{1}{\sqrt{x^2+2x+2}}$

$\cos(\tan^{-1}x)=\cos\left(\cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right)\right)=\frac{1}{\sqrt{1+x^2}}$

Thus $\frac{1}{\sqrt{1+(1+x)^2}}=\frac{1}{\sqrt{1+x^2}}$

$\Rightarrow x^2+2x+2=x^2+1 \Rightarrow x=-\frac{1}{2}$

86. (b)

87. (a) : Using $\tan^{-1}\theta+\cot^{-1}\theta=\pi/2=x$

$\therefore\ \sin x=\sin\pi/2=1$

88. (b)

89. (c) : We must have

$x(x+1)\geq 0$ ...(1)

and $0\leq x^2+x+1\leq 1$ ...(2)

From (1) $x^2+x+1\geq 1$

or $1\leq x^2+x+1\leq 1$ (from 2)

$\therefore\ x^2+x+1=1$

$\Rightarrow x(x+1)=0\ \therefore\ x=0$ or -1

The given equation is true for these two values. Hence the number of real solutions is two.

90. (d) : $\tan\left[\cos^{-1}\frac{1}{5\sqrt{2}}-\sin^{-1}\frac{4}{\sqrt{17}}\right]$

$=\tan[\tan^{-1}7-\tan^{-1}4]=\tan\left(\tan^{-1}\left(\frac{3}{29}\right)\right)=\frac{3}{29}$.

91. (e) : The principal value of

$\sin^{-1}\left(\sin\frac{2\pi}{3}\right)$

$=$ principal value of $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)=\frac{\pi}{3}$.

92. (b)

93. (a, b) : We have,

$S_n(x)=\sum_{k=1}^{n}\tan^{-1}\left(\frac{x}{1+k(k+1)x^2}\right)$

$=\sum_{k=1}^{n}\tan^{-1}\left(\frac{(kx+x)-(kx)}{1+(kx+x)(kx)}\right)$

$=\sum_{k=1}^{n}\tan^{-1}(k+1)x-\tan^{-1}kx$

$=\tan^{-1}(n+1)x-\tan^{-1}x$

$=\tan^{-1}\left(\frac{nx}{1+(n+1)x^2}\right)$

$\therefore\ S_{10}(x)=\tan^{-1}\left(\frac{10x}{1+11x^2}\right)$

$=\frac{\pi}{2}-\cot^{-1}\left(\frac{10x}{1+11x^2}\right)$

$=\frac{\pi}{2}-\tan^{-1}\left(\frac{1+11x^2}{10x}\right)\forall\ x>0$

So, (a) is true

Now, $\lim_{n\to\infty}\cot\left(\tan^{-1}\left(\frac{nx}{1+(n+1)x^2}\right)\right)$

$$=\lim_{n\to\infty}\frac{1+(n+1)x^2}{nx}=\lim_{n\to\infty}\left[\frac{1}{nx}+x+\frac{x}{n}\right]=x$$

So, (b) is true.

Now, $S_3(x)=\tan^{-1}4x-\tan^{-1}x=\frac{\pi}{4}$

$$\Rightarrow \tan^{-1}\frac{3x}{1+4x^2}=\frac{\pi}{4} \Rightarrow \frac{3x}{1+4x^2}=1$$

$$\Rightarrow 4x^2-3x+1=0$$

No real roots exist.

So, (c) is false.

$\tan(S_n(x))\to\frac{1}{x}$ when $n\to\infty$.

So, it doesn't have an upper bound for $x>0$.

94. (a, b, c) :

$$\sum_{k=0}^{n}\sin\left(\frac{k+1}{n+2}\pi\right)\sin\left(\frac{k+2}{n+2}\pi\right)$$

$$=\frac{1}{2}\sum_{k=0}^{n}\left\{\cos\left(\frac{\pi}{n+2}\right)-\cos\left(\frac{2k+3}{n+2}\right)\pi\right\}$$

$$=\frac{1}{2}\left[(n+1)\cos\frac{\pi}{n+2}-\frac{\sin\left(\frac{(n+1)}{n+2}\right)\pi}{\sin\frac{\pi}{n+2}}\cos\left(\frac{n+3}{n+2}\right)\pi\right]$$

$$=\frac{1}{2}\left[(n+1)\cos\left(\frac{\pi}{n+2}\right)+\cos\left(\frac{\pi}{n+2}\right)\right]$$

$$=\frac{1}{2}(n+2)\cos\left(\frac{\pi}{n+2}\right)$$

Again, $\sum_{k=0}^{n}\sin^2\left(\frac{k+1}{n+2}\right)\pi$

$$=\frac{1}{2}\sum_{k=0}^{n}\left[1-\cos\left(\frac{2k+2}{n+2}\right)\pi\right]$$

$$=\frac{1}{2}\left[(n+1)-\frac{\sin\left(\frac{(n+1)}{n+2}\right)\pi}{\sin\left(\frac{\pi}{n+2}\right)}\cdot\cos\frac{2(n+2)\pi}{2(n+2)}\right]$$

$$=\frac{1}{2}[n+1+1]=\frac{1}{2}(n+2)$$

Thus, $f(n)=\cos\left(\frac{\pi}{n+2}\right)$

$$\therefore\quad f(4)=\cos\frac{\pi}{6}=\frac{\sqrt{3}}{2}$$

$$\alpha=\tan(\cos^{-1}f(6))=\tan\left(\cos^{-1}\cos\left(\frac{\pi}{8}\right)\right)$$

$$=\tan\frac{\pi}{8}=\sqrt{2}-1$$

$$\therefore\quad \alpha(\alpha+2)=2-1\Rightarrow\alpha^2+2\alpha-1=0$$

$$\sin(7\cos^{-1}f(5))=\sin\left(7\cos^{-1}\cos\frac{\pi}{7}\right)$$

$$=\sin(\pi)=0$$

$$\lim_{n\to\infty}\cos\left(\frac{\pi}{n+2}\right)=1$$

95. (b, c, d) : $\alpha=3\sin^{-1}\frac{6}{11}>3\sin^{-1}\frac{6}{12}$

$$=\frac{3\pi}{6}=\frac{\pi}{2}\Rightarrow\quad\alpha>\frac{\pi}{2}$$

$\beta=3\cos^{-1}\frac{4}{9}>3\cos^{-1}\frac{4}{8}$ *i.e.*, $\beta>\pi$

Then, $\alpha+\beta>\frac{3\pi}{2}$

Also, $\alpha+\beta$ lies in fourth quadrant, hence $\cos(\alpha+\beta)>0$

96. (3) : We have,

$$y=\cos\left(\frac{\pi}{3}+\cos^{-1}\frac{x}{2}\right)$$

$$=\cos\frac{\pi}{3}\cos\left(\cos^{-1}\frac{x}{2}\right)-\sin\frac{\pi}{3}\sin\left(\cos^{-1}\frac{x}{2}\right)$$

$$=\frac{1}{2}\times\frac{x}{2}-\frac{\sqrt{3}}{2}\sin\left(\sin^{-1}\sqrt{1-\frac{x^2}{4}}\right)$$

$$=\frac{x}{4}-\frac{\sqrt{3}}{2}\frac{\sqrt{4-x^2}}{2}$$

$$\Rightarrow 4y=x-\sqrt{3(4-x^2)}$$

$$\Rightarrow (x-4y)=\sqrt{3(4-x^2)}$$

On squaring both sides, we get

$$x^2+16y^2-8xy=12-3x^2$$

$$\Rightarrow\quad 4x^2+16y^2-8xy=12$$

$$\Rightarrow\quad x^2+4y^2-2xy=3$$

$$\therefore\quad (x-y)^2+3y^2=3$$

97. (96) : We have,

$$f(x)=\cos^{-1}\left(\frac{4x+5}{3x-7}\right)$$

Since, domain of $\cos^{-1}x$ is $[-1, 1]$.

$$\therefore\quad -1\le\frac{4x+5}{3x-7}\le1$$

For, $\frac{4x+5}{3x-7}\ge-1$

$$\Rightarrow\frac{4x+5+3x-7}{3x-7}\ge0\Rightarrow\frac{7x-2}{3x-7}\ge0$$

$$\Rightarrow x\in\left(-\infty,\frac{2}{7}\right]\cup\left(\frac{7}{3},\infty\right)\qquad\text{...(i)}$$

Now, for $\frac{4x+5}{3x-7}\le1$

$$\Rightarrow\frac{4x+5-3x+7}{3x-7}\le0\Rightarrow\frac{x+12}{3x-7}\le0$$

$$\Rightarrow x\in\left[-12,\frac{7}{3}\right)\qquad\text{...(ii)}$$

On combining (i) and (ii), we get domain of $f(x)$ is $\left[-12,\frac{2}{7}\right]$.

$$\therefore\quad\alpha=-12\text{ and }\beta=\frac{2}{7}$$

Now, $g(x)=\log_2(2-6\log_{27}(2x+5))$

For $g(x)$ to be defined, we have

$2-6\log_{27}(2x+5)>0\Rightarrow 6\log_{27}(2x+5)<2$

$$\Rightarrow\quad\log_{27}(2x+5)<\frac{1}{3}$$

$$\Rightarrow\quad(2x+5)<(27)^{\frac{1}{3}}$$

$$\Rightarrow\quad 2x+5<3$$

$$\Rightarrow\quad 2x<-2\Rightarrow x<-1$$

Also, $2x+5>0$

[For $\log_{27}(2x+5)$ to be defined]

$$\Rightarrow x>\frac{-5}{2}$$

$\therefore$ Domain of $g(x)$ is $\left(\frac{-5}{2},-1\right)$

$$\therefore\quad\gamma=\frac{-5}{2}\text{ and }\delta=-1$$

$$\therefore\quad|7(\alpha+\beta)+4(\gamma+\delta)|$$

$$=\left|7\left(-12+\frac{2}{7}\right)+4\left(\frac{-5}{2}-1\right)\right|$$

$$=|-82-14|=96$$

98. (14) : Given, $\sec^2(\tan^{-1}\alpha)+\text{cosec}^2(\cot^{-1}\beta)=36$

$$\therefore\quad 1+\tan^2(\tan^{-1}\alpha)+1+\cot^2(\cot^{-1}\beta)=36$$

$$\Rightarrow\quad\alpha^2+\beta^2=34$$

Given, $\alpha+\beta=8$ ....(i)

$$\Rightarrow\quad\alpha^2+\beta^2+2\alpha\beta=64$$

$$\Rightarrow\quad 2\alpha\beta=30\Rightarrow\alpha\beta=15$$

$\therefore\ \beta-\alpha=\sqrt{(\alpha+\beta)^2-4\alpha\beta}\quad[\because\beta\ge\alpha]$

$\beta-\alpha=2$ ... (ii)

From (i) and (ii), we get

$\alpha = 3, \beta = 5$

$\therefore \alpha^2 + \beta = 9 + 5 = 14$

99. (5) : We have,

$$\cos^{-1}x = \pi + \sin^{-1}x + \sin^{-1}(2x+1)$$

$$\Rightarrow 2\cos^{-1}x - \sin^{-1}(2x+1) = \frac{3\pi}{2}$$

$$\Rightarrow 2\alpha - \beta = \frac{3\pi}{2}, \text{where } \cos^{-1}x = \alpha, \sin^{-1}(2x+1) = \beta$$

$$\Rightarrow 2\alpha = \frac{3\pi}{2} + \beta \Rightarrow \cos 2\alpha = \sin\beta$$

$$\Rightarrow 2\cos^2\alpha - 1 = \sin\beta \Rightarrow 2x^2 - 1 = 2x + 1$$

[$\because x = \cos\alpha$ and $2x + 1 = \sin\beta$]

$$\Rightarrow x^2 - x - 1 = 0$$

$$\Rightarrow x = \frac{1 \pm \sqrt{5}}{2} \Rightarrow x = \frac{1-\sqrt{5}}{2}$$

$$(\because x = \frac{1+\sqrt{5}}{2} \text{ rejected as} \cos\alpha \le 1)$$

Now, $\sum_{x \in S}(2x-1)^2 = \sum_{x \in S}(4x^2 + 1 - 4x) = 5$

100. (47) : $\cot^{-1}3 + \cot^{-1}4 + \cot^{-1}5 + \cot^{-1}n = \frac{\pi}{4}$

$$\Rightarrow \tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{4} + \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{n} = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1}\left(\frac{\frac{1}{3}+\frac{1}{4}}{1-\frac{1}{12}}\right) + \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{n} = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1}\left(\frac{7}{11}\right) + \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{n} = \tan^{-1}1$$

$$\Rightarrow \tan^{-1}\left(\frac{\left(\frac{7}{11}+\frac{1}{5}\right)}{1-\frac{7}{11}\cdot\frac{1}{5}}\right) + \tan^{-1}\frac{1}{n} = \tan^{-1}1$$

$$\Rightarrow \tan^{-1}\left(\frac{23}{24}\right) + \tan^{-1}\frac{1}{n} = \tan^{-1}1$$

$$\Rightarrow \tan^{-1}\frac{1}{n} = \tan^{-1}1 - \tan^{-1}\frac{23}{24}$$

$$\Rightarrow \tan^{-1}\frac{1}{n} = \tan^{-1}\left(\frac{1-\frac{23}{24}}{1+\frac{23}{24}}\right)$$

$$\Rightarrow \frac{1}{n} = \frac{1}{47} \Rightarrow n = 47$$

101. (0) : We have, $2\sin^{-1}x + 3\cos^{-1}x = \frac{2\pi}{5}$

$$\Rightarrow 2\left(\frac{\pi}{2} - \cos^{-1}x\right) + 3\cos^{-1}x = \frac{2\pi}{5}$$

$$\Rightarrow \pi + \cos^{-1}x = \frac{2\pi}{5}$$

$$\Rightarrow \cos^{-1}x = \frac{-3\pi}{5}$$

Which is not possible as $\cos^{-1}x \in [0, \pi]$

$\therefore$ No solution exist.

102. (24) : Given $f(x) = \sec^{-1}\left(\frac{2x}{5x+3}\right)$

So, $\frac{2x}{5x+3} \ge 1$

$$\Rightarrow \frac{2x}{5x+3} - 1 \ge 0 \Rightarrow \frac{-3x-3}{5x+3} \ge 0$$

$$\Rightarrow \frac{x+1}{5x+3} \le 0 \Rightarrow x \in \left[-1, \frac{-3}{5}\right)$$

Now, $\frac{2x}{5x+3} \le -1 \Rightarrow \frac{2x}{5x+3} + 1 \le 0$

$$\Rightarrow \frac{7x+3}{5x+3} \le 0 \Rightarrow x \in \left(\frac{-3}{5}, \frac{-3}{7}\right]$$

$$\therefore x \in \left[-1, \frac{-3}{5}\right) \cup \left(\frac{-3}{5}, \frac{-3}{7}\right]$$

$$\alpha = -1, \beta = \frac{-3}{5}, \gamma = \frac{-3}{5}, \delta = \frac{-3}{7}$$

So, $|3\alpha + 10(\beta+\gamma) + 21\delta|$

$$= \left|3\times(-1) + 10\left(\frac{-3}{5} - \frac{3}{5}\right) + 21\times\left(\frac{-3}{7}\right)\right|$$

$= |-3 -12-9| = 24$

103. (4) : Given,

$$\sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right) - \sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right) = \frac{\pi}{4}$$

$$\Rightarrow \sin^{-1}\left(\frac{x+1}{\sqrt{(x+1)^2+1}}\right) - \sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right) = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1}(x+1) - \tan^{-1}x = \frac{\pi}{4}$$

$$\Rightarrow \tan^{-1}\left[\frac{(x+1)-x}{1+(x+1)x}\right] = \frac{\pi}{4}$$

$$\Rightarrow \frac{1}{x^2+x+1} = \tan\left(\frac{\pi}{4}\right) = 1$$

$$\Rightarrow x^2 + x + 1 = 1 \Rightarrow x^2 + x = 0$$

$\Rightarrow x = 0$ or -1 $\quad \therefore S$ is $\{-1, 0\}$

$$\therefore \sum_{x \in s}\sin\left((x^2+x+5)\frac{\pi}{2}\right) - \cos((x^2+x+5)\pi)$$

$$= \left[\sin\left(\frac{5\pi}{2}\right) - \cos(5\pi)\right] + \left\{\sin\left(\frac{5\pi}{2}\right) - \cos(5\pi)\right\}$$

$= 2 + 2 = 4$

104. (2) : $\sin^{-1}x = 2\tan^{-1}x$

$$\sin^{-1}x = \sin^{-1}\left(\frac{2x}{1+x^2}\right) \Rightarrow x = \frac{2x}{1+x^2}$$

$$\Rightarrow x(1+x^2) = 2x$$

$\Rightarrow x(1 + x^2) - 2x = 0 \Rightarrow x(x^2 - 1) = 0$

$\Rightarrow x = 0$ or $x^2 - 1 = 0$; $x^2 = 1$; $x = \pm1$

$\Rightarrow x = 0$ or $x = 1, -1$

No. of solutions = 2 $\quad \because x = -1 \notin (-1, 1]$

105. (2) : Given,

$$\tan^{-1}\left(\frac{2x}{1-x^2}\right) + \cot^{-1}\left(\frac{1-x^2}{2x}\right) = \frac{\pi}{3}$$

Case I : If $x > 0$

$$\tan^{-1}\left(\frac{2x}{1-x^2}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \frac{\pi}{3}$$

$$\Rightarrow \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \frac{\pi}{6}$$

$$\Rightarrow \frac{2x}{1-x^2} = \frac{1}{\sqrt{3}} \Rightarrow x^2 + 2\sqrt{3}x - 1 = 0$$

$$\Rightarrow [x-(2-\sqrt{3})][x+(2+\sqrt{3})] = 0$$

$$\Rightarrow x = 2-\sqrt{3}, -(2+\sqrt{3})$$

$x = -(2+\sqrt{3})$ is rejected because $x > 0$.

Hence, $x = 2-\sqrt{3}$

Case II : If $x < 0$

$$\tan^{-1}\left(\frac{2x}{1-x^2}\right) + \tan^{-1}\left(\frac{2x}{1-x^2}\right) = \frac{\pi}{3} - \pi = -\frac{2\pi}{3}$$

$$\Rightarrow \tan^{-1}\left(\frac{2x}{1-x^2}\right) = -\frac{\pi}{3} \Rightarrow \frac{2x}{1-x^2} = -\sqrt{3}$$

$$\Rightarrow \sqrt{3}x^2 - 2x - \sqrt{3} = 0$$

$$\Rightarrow (\sqrt{3}x - 3)\left(x + \frac{1}{\sqrt{3}}\right) = 0$$

$$\Rightarrow x = \frac{-1}{\sqrt{3}}, \sqrt{3}$$

$\Rightarrow x = \sqrt{3}$ is rejected because $x < 0$.

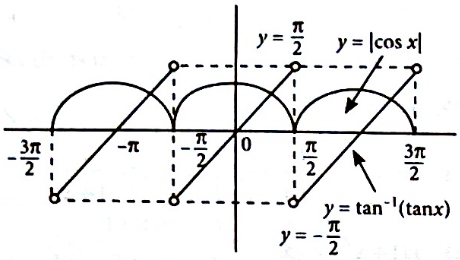
Thus $x = -\frac{1}{\sqrt{3}}$

$\therefore$ Sum of solution

$$= 2 - \sqrt{3} - \frac{1}{\sqrt{3}} = 2 - \frac{4}{\sqrt{3}}$$

$$\alpha - \frac{4}{\sqrt{3}} = 2 - \frac{4}{\sqrt{3}} \Rightarrow \alpha = 2$$

106. (3) : $\sqrt{1+\cos(2x)} = \sqrt{2}\tan^{-1}(\tan x)$



$\Rightarrow \quad \sqrt{2}|\cos x| = \sqrt{2}\tan^{-1}(\tan x)$

$\Rightarrow \quad |\cos x| = \tan^{-1}(\tan x)$

Number of solution = Number of intersection points = 3

107. (130) : $x = \sin(2\tan^{-1}\alpha)$

and $y = \sin\left(\frac{1}{2}\tan^{-1}\frac{4}{3}\right)$

$x = \sin(2\theta),\ \theta = \tan^{-1}\alpha$

$\Rightarrow \quad x = \frac{2\tan\theta}{1+\tan^2\theta},\ \alpha = \tan\theta$

$\Rightarrow \quad x = \frac{2\alpha}{1+\alpha^2}$ ...(i)

and $y = \sin\left(\frac{1}{2}\tan^{-1}\left(\frac{4}{3}\right)\right)$

$y = \sin\theta,\ \frac{1}{2}\tan^{-1}\frac{4}{3} = \theta$

$\Rightarrow \quad y = \sin\theta,\ \tan 2\theta = \frac{4}{3}$

$\Rightarrow \quad y = \sin\theta,\ \tan\theta = \frac{1}{2} \Rightarrow \quad y = \frac{1}{\sqrt{5}}$

Now, $y^2 = 1 - x$

$\frac{1}{5} = 1 - \frac{2\alpha}{1+\alpha^2} \Rightarrow \frac{1}{5} = \frac{1+\alpha^2-2\alpha}{1+\alpha^2}$

$\Rightarrow \quad 2\alpha^2 - 5\alpha + 2 = 0 \Rightarrow \alpha = \frac{1}{2}, 2$

$\sum_{\alpha\in S} 16\alpha^3 = 16\times(2)^3 + 16\times\left(\frac{1}{2}\right)^3$

$= 128 + 2 = 130$

108. (12) : $\cos(\sin^{-1}(x\cot(\tan^{-1}(\cos(\sin^{-1}x))))) = k$

$\Rightarrow \quad \cos(\sin^{-1}(x\cot(\tan^{-1}(\cos(\cos^{-1}\sqrt{1-x^2}))))) = k$

$\Rightarrow \cos(\sin^{-1}(x\cot(\tan^{-1}\sqrt{1-x^2}))) = k$

$\Rightarrow \cos(\sin^{-1}(x\cot(\cot^{-1}\frac{1}{\sqrt{1-x^2}}))) = k$

$\Rightarrow \cos\left(\sin^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)\right) = k$

$\Rightarrow \cos\left(\cos^{-1}\left(\frac{\sqrt{1-2x^2}}{\sqrt{1-x^2}}\right)\right) = k$

$\Rightarrow \frac{\sqrt{1-2x^2}}{\sqrt{1-x^2}} = k \Rightarrow 1 - 2x^2 = k^2(1-x^2)$

$\Rightarrow 1 - 2x^2 = k^2 - k^2x^2 \Rightarrow (k^2-2)x^2 + (1-k^2) = 0$

Given α, β are solution then, $\alpha + \beta = 0,\ \alpha = -\beta$

$\alpha\cdot\beta = \frac{1-k^2}{k^2-2},\ \beta^2 = \frac{k^2-1}{k^2-2}$

Given, solution of equation $x^2 - bx - 5$ are $\left(\frac{1}{\alpha^2}+\frac{1}{\beta^2}\right)$ and $\left(\frac{\alpha}{\beta}\right)$

$\frac{1}{\alpha^2}+\frac{1}{\beta^2}+\frac{\alpha}{\beta} = b$ ...(i)

$\left(\frac{1}{\alpha^2}+\frac{1}{\beta^2}\right)\left(\frac{\alpha}{\beta}\right) = -5$ ...(ii)

Equation (ii) can be written as,

$\left(\frac{1}{\beta^2}+\frac{1}{\beta^2}\right)(-1) = -5 \Rightarrow \frac{2}{\beta^2} = 5 \Rightarrow \beta^2 = \frac{2}{5}$

Put $\beta^2 = \frac{2}{5}$ and $\alpha = -\beta$ in equation (i)

$\Rightarrow \quad \frac{1}{\beta^2}+\frac{1}{\beta^2} - 1 = b \Rightarrow b = 4$

Put $\beta^2 = \frac{2}{5}$ in $\beta^2 = \frac{k^2-1}{k^2-2} \quad \Rightarrow \quad k^2 = \frac{1}{3}$

Now, $\frac{b}{k^2} = \frac{4}{\left(\frac{1}{3}\right)} = 12$

109. (29) :

$50\tan\left(3\tan^{-1}\left(\frac{1}{2}\right)+2\cos^{-1}\left(\frac{1}{\sqrt{5}}\right)\right) + 4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}2\sqrt{2}\right)$

$= 50\tan\left(3\tan^{-1}\left(\frac{1}{2}\right)+2\tan^{-1}2\right) + 4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}2\sqrt{2}\right)$

$= 50\tan\left(2\cot^{-1}2+2\tan^{-1}2+\tan^{-1}\frac{1}{2}\right) + 4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}2\sqrt{2}\right)$

$= 50\tan\left(\pi+\tan^{-1}\frac{1}{2}\right) + 4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}2\sqrt{2}\right)$

$= 50\tan\left(\tan^{-1}\frac{1}{2}\right) + 4\sqrt{2}\tan\left(\frac{1}{2}\tan^{-1}2\sqrt{2}\right)$

$= 25 + 4\sqrt{2}\tan\alpha$ ...(i)

where, $\alpha = \frac{1}{2}\tan^{-1}2\sqrt{2}$

$\Rightarrow \quad \tan 2\alpha = 2\sqrt{2} \Rightarrow \frac{2\tan\alpha}{1-\tan^2\alpha} = 2\sqrt{2}$

$\Rightarrow \quad 2\sqrt{2}\tan^2\alpha + 2\tan\alpha - 2\sqrt{2} = 0$

$\Rightarrow \quad \sqrt{2}\tan^2\alpha + 2\tan\alpha - \tan\alpha - \sqrt{2} = 0$

$\Rightarrow \quad \sqrt{2}\tan\alpha(\tan\alpha+\sqrt{2}) - 1(\tan\alpha+\sqrt{2}) = 0$

$\Rightarrow \quad (\sqrt{2}\tan\alpha - 1)(\tan\alpha+\sqrt{2}) = 0$

$\Rightarrow \quad \tan\alpha = \frac{1}{\sqrt{2}}$ or $\tan\alpha = -\sqrt{2}$

Since, $2\alpha \in \left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ *i.e.*, $\alpha \in \left(\frac{-\pi}{4}, \frac{\pi}{4}\right)$

$\Rightarrow \quad \tan\alpha = \frac{1}{\sqrt{2}}$

Substituting the value of $\tan\alpha$ in equation (i), we get $25 + 4\sqrt{2}\cdot\frac{1}{\sqrt{2}} = 29.$

110. (2.36) : $\cos^{-1}\sqrt{\frac{2}{2+\pi^2}} = \tan^{-1}\frac{\pi}{\sqrt{2}}$

Also, $\sin^{-1}\left(\frac{2\sqrt{2}\pi}{2+\pi^2}\right) = \sin^{-1}\left(\frac{2\pi/\sqrt{2}}{1+(\pi/\sqrt{2})^2}\right)$

$= \pi - 2\tan^{-1}\frac{\pi}{\sqrt{2}} \quad \left(\text{As } \frac{\pi}{\sqrt{2}} > 1\right)$

$\left[\text{Recall that } \sin^{-1}\frac{2x}{1+x^2} = \pi - 2\tan^{-1}x, \text{ for } x \geq 1\right]$

Now $\frac{3}{2}\cos^{-1}\sqrt{\frac{2}{2+\pi^2}} + \frac{1}{4}\sin^{-1}\frac{2\sqrt{2}\pi}{2+\pi^2} + \tan^{-1}\frac{\sqrt{2}}{\pi}$

$= \frac{3}{2}\cdot\tan^{-1}\frac{\pi}{\sqrt{2}} + \frac{1}{4}\left(\pi - 2\tan^{-1}\frac{\pi}{\sqrt{2}}\right) + \tan^{-1}\frac{\sqrt{2}}{\pi}$

$= \tan^{-1}\frac{\pi}{\sqrt{2}} + \tan^{-1}\frac{\sqrt{2}}{\pi} + \frac{\pi}{4}$

$= \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4} = 2.36$ (Using $\pi = 3.141$)

111. (1) :

$\lim_{n\to\infty}\tan\left(\sum_{r=1}^{n}\tan^{-1}\left(\frac{1}{1+r(r+1)}\right)\right)$

$= \lim_{n\to\infty}\tan\left(\sum_{r=1}^{n}\tan^{-1}\left(\frac{r+1-r}{1+r(r+1)}\right)\right)$

$= \tan\left(\lim_{n\to\infty}\sum_{r=1}^{n}[\tan^{-1}(r+1) - \tan^{-1}(r)]\right)$

$= \tan\left(\lim_{n\to\infty}\left(\tan^{-1}(n+1) - \tan^{-1}1\right)\right)$

$$=\tan\left(\tan^{-1}(\infty)-\frac{\pi}{4}\right)=\tan\left(\frac{\pi}{2}-\frac{\pi}{4}\right)=\tan\left(\frac{\pi}{4}\right)=1$$

112. (0) :

Consider $\displaystyle\sum_{k=0}^{10}\frac{1}{\cos\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\cos\left(\frac{7\pi}{12}+\frac{(k+1)\pi}{2}\right)}$

$$=\frac{1}{\sin\frac{\pi}{2}}\sum_{k=0}^{10}\frac{\sin\left\{\left(\frac{7\pi}{12}+\frac{(k+1)\pi}{2}\right)-\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\right\}}{\cos\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\cos\left(\frac{7\pi}{12}+\frac{(k+1)\pi}{2}\right)}$$

$$=\sum_{k=0}^{10}\left[\tan\left(\frac{7\pi}{12}+(k+1)\frac{\pi}{2}\right)-\tan\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\right]$$

$$=\tan\left(\frac{11\pi}{2}+\frac{7\pi}{12}\right)-\tan\frac{7\pi}{12}$$

$$=-\cot\frac{7\pi}{12}-\tan\frac{7\pi}{12}$$

$$=-\frac{1}{\sin\frac{7\pi}{12}\cos\frac{7\pi}{12}}=-\frac{2}{\sin\frac{7\pi}{6}}=\frac{2}{\sin\frac{\pi}{6}}=4$$

$$\therefore \sec^{-1}\left(\frac{1}{4}\sum_{k=0}^{10}\sec\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\sec\left(\frac{7\pi}{12}+\frac{(k+1)\pi}{2}\right)\right)$$

$$=\sec^{-1}(1)=0$$

113. (2) : Let $f(x)=\displaystyle\sum_{i=1}^{\infty}x^{i+1}-x\sum_{i=1}^{\infty}\left(\frac{x}{2}\right)^i$

$$=(x^2+x^3+\ldots\text{ to }\infty)-x\left(\frac{x}{2}+\left(\frac{x}{2}\right)^2+\ldots\text{ to }\infty\right)$$

$$=\frac{x^2}{1-x}-\frac{x\cdot\frac{x}{2}}{1-\frac{x}{2}}=\frac{x^2}{1-x}-\frac{x^2}{2-x}$$

Again, let $g(x)=\displaystyle\sum_{i=1}^{\infty}\left(-\frac{x}{2}\right)^i-\sum_{i=1}^{\infty}(-x)^i$

$$=\left[\left(\frac{-x}{2}\right)+\left(\frac{-x}{2}\right)^2+\ldots\text{ to }\infty\right]-\left[(-x)+(-x)^2+\ldots\text{ to }\infty\right]$$

$$=\frac{-\frac{x}{2}}{1+\frac{x}{2}}-\frac{-x}{1+x}=\frac{x}{1+x}-\frac{x}{2+x}$$

$\therefore$ The given equation becomes

$\sin^{-1}f(x)+\cos^{-1}g(x)=\pi/2$

So, we must have $f(x)=g(x)$

$$\Rightarrow \frac{x^2}{1-x}-\frac{x^2}{2-x}=\frac{x}{1+x}-\frac{x}{2+x}$$

$$\Rightarrow \frac{x^2}{(1-x)(2-x)}=\frac{x}{(2+x)(1+x)}$$

$\Rightarrow$ $x=0$ and $x(2+x)(1+x)=(1-x)(2-x)$

Let $h(x)=x(2+x)(1+x)-(1-x)(2-x)$

$=x^3+2x^2+5x-2$

Now, $h\left(-\frac{1}{2}\right)=\left(-\frac{1}{2}\right)^3+2\left(-\frac{1}{2}\right)^2+5\left(-\frac{1}{2}\right)-2$

$$=-\frac{1}{8}+\frac{1}{2}-\frac{5}{2}-2=-\frac{33}{8}<0$$

$$h\left(\frac{1}{2}\right)=\left(\frac{1}{2}\right)^3+2\left(\frac{1}{2}\right)^2+5\left(\frac{1}{2}\right)-2$$

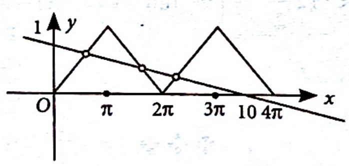
$$=\frac{1}{8}+\frac{1}{2}+\frac{5}{2}-2=\frac{9}{8}>0$$

$\therefore$ There exists a root between $-\frac{1}{2}$ and $\frac{1}{2}$.

Also, $h'(x)=3x^2+4x+5>0$

$\Rightarrow$ $h(x)$ has exactly one real root in $\left(-\frac{1}{2},\frac{1}{2}\right)$.

114. (3) : The graph of $y=\cos^{-1}(\cos x)$ and $y=\frac{10-x}{10}=1-\frac{x}{10}$ is shown.



The number of solutions is the number of points of intersection *i.e.*, 3

115. (1) : $\tan^{-1}\left(\frac{\sin\theta}{\sqrt{\cos 2\theta}}\right)$

$$=\sin^{-1}\left(\frac{\sin\theta}{\sqrt{\sin^2\theta+\cos 2\theta}}\right)$$

$$=\sin^{-1}\left(\frac{\sin\theta}{\sqrt{\cos^2\theta}}\right)=\sin^{-1}\left(\frac{\sin\theta}{\cos\theta}\right)$$

as $-\frac{\pi}{4}<\theta<\frac{\pi}{4}$

Thus $f(\theta)=\sin\left(\tan^{-1}\left(\frac{\sin\theta}{\sqrt{\cos^2\theta}}\right)\right)$

$$=\sin\left(\sin^{-1}\left(\frac{\sin\theta}{\cos\theta}\right)\right)$$

$=\sin(\sin^{-1}(\tan\theta))=\tan\theta$

Now, $\frac{d}{d(\tan\theta)}(f(\theta))=1$

116. (b) :

(P) $\left[\frac{1}{y^2}\left(\frac{\cos(\tan^{-1}y)+y\sin(\tan^{-1}y)}{\cot(\sin^{-1}y)+\tan(\sin^{-1}y)}\right)^2+y^4\right]^{1/2}$

$$=\frac{1}{y^2}\frac{\left\{\frac{1}{\sqrt{1+y^2}}+\frac{y^2}{\sqrt{1+y^2}}\right\}^2}{\left(\frac{\sqrt{1-y^2}}{y}+\frac{y}{\sqrt{1-y^2}}\right)^2}+y^4$$

$$=\left[\frac{1}{y^2}\cdot y^2(1-y^4)+y^4\right]^{1/2}=1$$

(Q) $\cos x+\cos y=-\cos z$ and $\sin x+\sin y=-\sin z$

Squaring and adding we get

$2+2\cos(x-y)=1$ $\therefore$ $\cos(x-y)=-1/2$

$$2\cos^2\left(\frac{x-y}{2}\right)-1=-\frac{1}{2}\ \therefore\ \cos\left(\frac{x-y}{2}\right)=\frac{1}{2}$$

(R) $\cos 2x\left\{\cos\left(\frac{\pi}{4}-x\right)-\cos\left(\frac{\pi}{4}+x\right)\right\}+2\sin^2 x$

$=2\sin x\cos x$

$=\cos 2x(\sqrt{2}\sin x)+2\sin^2 x=2\sin x\cos x$

$=\sqrt{2}\sin x[\cos 2x+\sqrt{2}\sin x-\sqrt{2}\cos x]=0$

$\Rightarrow$ Either $\sin x=0$ or $\cos^2 x-\sin^2 x=\sqrt{2}(\cos x-\sin x)$

$\Rightarrow$ $\sec x=1$ or $\cos x=\sin x$

$\Rightarrow$ $\sec x=1$ or $\sec x=\operatorname{cosec} x$

(S) $\cot(\sin^{-1}\sqrt{1-x^2})=\sin(\tan^{-1}(x\sqrt{6}))$

$\Rightarrow$ $\frac{|x|}{\sqrt{1-x^2}}=\frac{\sqrt{6}x}{\sqrt{1+6x^2}}$, $x>0$

$\Rightarrow$ $1+6x^2=6-6x^2$

$\Rightarrow$ $12x^2=5$ $\therefore$ $x=\frac{\sqrt{5}}{2\sqrt{3}}$

117. (i) → (A), (ii) → (B), (iii) → (A), (iv) → (D)

$\sin^{-1}(ax)+\cos^{-1}(y)+\cos^{-1}(bxy)=\pi/2$

or $\cos^{-1}y+\cos^{-1}(bxy)=\frac{\pi}{2}-\sin^{-1}(ax)=\cos^{-1}(ax)$

or $\cos^{-1}(bxy)=\cos^{-1}(ax)-\cos^{-1}(y)$

or $\cos^{-1}(bxy)=\cos^{-1}\left(axy+\sqrt{1-a^2x^2}\sqrt{1-y^2}\right)$

$\Rightarrow$ $bxy=axy+\sqrt{1-a^2x^2}\sqrt{1-y^2}$

$\Rightarrow$ $(b-a)xy=\sqrt{1-a^2x^2}\sqrt{1-y^2}$

or $(b-a)^2x^2y^2=(1-a^2x^2)(1-y^2)$

or $a^2x^2+y^2+(b-a)^2x^2y^2-a^2x^2y^2-1=0$

or $a^2x^2+y^2+(b^2-2ab)x^2y^2=1$.

(i) If $a=1$, $b=0$, $x^2+y^2=1$

(ii) If $a = 1, b = 1$

$x^2 + y^2 - x^2y^2 - 1 = 0, (x^2 - 1)(y^2 - 1) = 0$

(iii) If $a = 1, b = 2, x^2 + y^2 = 1$

(iv) $a = 2, b = 2, 4x^2 + y^2 - 4x^2y^2 - 1 = 0$

$(4x^2 - 1)(y^2 - 1) = 0$

118. L.H.S. $= \cos(\tan^{-1} \sin \cot^{-1} x)$

$= \cos(\tan^{-1} \sin \alpha)$ where $\cot \alpha = x$

$$= \cos\left(\tan^{-1} \frac{1}{\sqrt{1+x^2}}\right) \quad \left[\because \sin\alpha = \frac{1}{\sqrt{1+x^2}}\right]$$

$= \cos\beta$ where $\tan\beta = \dfrac{1}{\sqrt{1+x^2}}$

$$= \sqrt{\frac{1+x^2}{2+x^2}}.$$

119. $\cos(2\cos^{-1}x + \sin^{-1}x)$

$$= \cos\left(\frac{\pi}{2} + \cos^{-1} x\right)$$

$$\left(\text{Using } \cos^{-1} x + \sin^{-1} x = \frac{\pi}{2}\right)$$

$$= -\sin(\cos^{-1} x) = -\sqrt{1-x^2} = -\frac{\sqrt{24}}{5}$$

120. $\tan^{-1}\left(\dfrac{2x+3x}{1-6x^2}\right) = \dfrac{\pi}{4} \quad \therefore \quad \dfrac{5x}{1-6x^2} = 1$

$\Rightarrow \quad 6x^2 + 5x - 1 = 0$

$\Rightarrow \quad (x + 1)(6x - 1) = 0$

$\therefore \quad x = -1$ or $\dfrac{1}{6}$

$x = -1$ does not satisfy the equation where as $x = 1/6$ satisfies.

$x = 1/6$

121. $A = 2\tan^{-1}(2\sqrt{2} - 1) > 2\tan^{-1}\sqrt{3}$

as $2\sqrt{2} - 1 > \sqrt{3}$

i.e., $A > 120°$

Now $B = 3\sin^{-1}\dfrac{1}{3} + \sin^{-1}\dfrac{3}{5}$

$$= 3\tan^{-1}\frac{1}{2\sqrt{2}} + \tan^{-1}\frac{3}{4}$$

$$< 3\tan^{-1}(\sqrt{2} - 1) + \tan^{-1}(1)$$

$\therefore \quad B < 112\dfrac{1°}{2} \qquad \left(\because \ \tan^{-1}(\sqrt{2} - 1) = 22\dfrac{1°}{2}\right)$

$\therefore \quad A > B.$

122. $2\tan^{-1}\dfrac{1}{5} = \tan^{-1}\left(\dfrac{2/5}{1-\frac{1}{25}}\right) = \tan^{-1}\left(\dfrac{5}{12}\right)$

$$\tan\left(2\tan^{-1}\frac{1}{5} - \frac{\pi}{4}\right) = \tan\left(\tan^{-1}\frac{5}{12} - \frac{\pi}{4}\right)$$

$$= \frac{\frac{5}{12} - 1}{1 + \frac{5}{12}} = -\frac{7}{17}.$$

123. Let $x = \sqrt{\dfrac{a(a+b+c)}{bc}}$,

$y = \sqrt{\dfrac{b(a+b+c)}{ca}}$ and $z = \sqrt{\dfrac{c(a+b+c)}{ab}}$

clearly $x > 0, y > 0$, $xy = \dfrac{a+b+c}{c} > 1$

$$\therefore \quad \tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)$$

$$= \pi + \tan^{-1}\left(\frac{\sqrt{\frac{a+b+c}{c}}\left(\sqrt{\frac{a}{b}} + \sqrt{\frac{b}{a}}\right)}{1 - \frac{a+b+c}{c}}\right)$$

$$= \pi - \tan^{-1}\left(\sqrt{\frac{c(a+b+c)}{ab}}\right) = \pi - \tan^{-1} z$$

$\therefore \quad \theta = \tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$

$\Rightarrow \quad \tan\theta = \tan\pi = 0$